From First Principles to Modern Large Language Models

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Contents

Block A — Foundations

Chapter 1: Sets, functions, logic, proofs

Motivation

Every object in this book—a vector, a probability distribution, a neural network, a token—is ultimately a *set with structure*. Before we can build the embedding map of Chapter 19 or even define the real numbers in Chapter 5, we need a precise grammar for membership, functions, and proof. Chapter 8 (linear maps) treats functions between vector spaces; Chapter 15 (probability) treats measures on σ-algebras of sets. Both collapse without the vocabulary below. We therefore start, with no apology, from the bottom.

Definitions

Definition 0.1 (Set, membership, subset)**.** *A* set *S is a collection of distinct objects called elements. We write $x \in S$ when x is an element of S, and $x \notin S$ otherwise. The* empty set *$\emptyset$ is the unique set with no elements. We say A is a* subset *of B, written $A \subset B$, iff $\forall x\,(x \in A \Rightarrow x \in B)$. Two sets are equal, $A = B$, iff $A \subset B$ and $B \subset A$.*

Definition 0.2 (Boolean operations)**.** *Given A, B inside a universe U:*

$$\begin{aligned} A \cup B &:= \{x \in U : x \in A \vee x \in B\}, \\ A \cap B &:= \{x \in U : x \in A \wedge x \in B\}, \\ A^c &:= \{x \in U : x \notin A\}, \\ \mathcal{P}(S) &:= \{T : T \subset S\}. \end{aligned}$$

Definition 0.3 (Function, image, preimage)**.** *A* function *$f : A \to B$ assigns to each $a \in A$ exactly one $f(a) \in B$. The image is $f(A) := \{f(a) : a \in A\}$ and the preimage of $T \subset B$ is $f^{-1}(T) := \{a \in A : f(a) \in T\}$. We call f:*

- injective *iff $\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2$;*
- surjective *iff $\forall b \in B,\ \exists a \in A,\ f(a) = b$;*
- bijective *iff both.*

Remark 0.4 (Logic). Propositional connectives: $\wedge, \vee, \neg, \Rightarrow, \Leftrightarrow$. Quantifiers: $\forall, \exists$. Standard proof techniques: *direct* (assume P, derive Q), *contrapositive* ($P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$), *contradiction* (assume $\neg P$, derive $\perp$), and *induction* (Theorem 0.7).

Theorems and proofs

Theorem 0.5 (De Morgan's laws for sets)**.** *For sets $A, B \subset U$,*

$$(A \cup B)^c = A^c \cap B^c \qquad \textit{and} \qquad (A \cap B)^c = A^c \cup B^c.$$

Proof. We prove the first; the second is symmetric. Fix arbitrary $x \in U$. We show the biconditional $x \in (A \cup B)^c \Leftrightarrow x \in A^c \cap B^c$.

($\Rightarrow$) Assume $x \in (A \cup B)^c$. By definition, $x \notin A \cup B$, i.e. $\neg(x \in A \vee x \in B)$. By the propositional De Morgan law $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$, this is $\neg(x \in A) \wedge \neg(x \in B)$, i.e. $x \notin A$ and $x \notin B$. Hence $x \in A^c$ and $x \in B^c$, so $x \in A^c \cap B^c$.

($\Leftarrow$) Assume $x \in A^c \cap B^c$. Then $x \notin A$ and $x \notin B$, so $\neg(x \in A) \wedge \neg(x \in B)$, which by propositional De Morgan equals $\neg(x \in A \vee x \in B)$, i.e. $x \notin A \cup B$, i.e. $x \in (A \cup B)^c$.

For the second identity, the same argument with $\wedge$ and $\vee$ interchanged (using $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$) gives $x \in (A \cap B)^c \Leftrightarrow x \in A^c \cup B^c$. □

Theorem 0.6 (Composition of injections is injective)**.** *If $f : A \to B$ and $g : B \to C$ are injective, then $g \circ f : A \to C$ is injective.*

Proof. Let $a_1, a_2 \in A$ with $(g \circ f)(a_1) = (g \circ f)(a_2)$, i.e. $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. By definition $g \circ f$ is injective. □

Theorem 0.7 (Principle of mathematical induction)**.** *Let $P(n)$ be a predicate on $\mathbb{N} = \{1, 2, 3, \ldots\}$. If*

(i) $P(1)$ *holds, and*

(ii) $\forall n \in \mathbb{N},\ P(n) \Rightarrow P(n+1)$,

then $\forall n \in \mathbb{N},\ P(n)$.

Proof. We assume the well-ordering principle: every nonempty subset of $\mathbb{N}$ has a least element. Let

$$S := \{n \in \mathbb{N} : \neg P(n)\}.$$

We show $S = \emptyset$ by contradiction. Suppose $S \neq \emptyset$. By well-ordering, S has a least element m. By (i), $P(1)$ holds, so $1 \notin S$, hence $m \geq 2$, so $m-1 \in \mathbb{N}$. By minimality of m in S, $m-1 \notin S$, i.e. $P(m-1)$ holds. By (ii) applied at $n = m-1$, $P(m)$ holds, so $m \notin S$. This contradicts $m \in S$. Therefore $S = \emptyset$, i.e. $P(n)$ holds for every $n \in \mathbb{N}$. □

Code sketch

In the companion notebook we (a) enumerate $\mathcal{P}(\{a,b,c\})$ from scratch and verify $|\mathcal{P}(S)| = 2^{|S|}$; (b) brute-force De Morgan on $U = \{1,\ldots,8\}$ with $A = \{1,3,5,7\}$, $B = \{2,3,5,7\}$; (c) implement `is_injective`, `is_surjective`, `is_bijective` as predicates over finite functions encoded as Python dictionaries; (d) verify $\sum_{k=1}^{n} k = n(n+1)/2$ both by direct summation up to $n = 20$ and by checking the induction step $S(n+1) - S(n) = n+1$ numerically.

Connection to LLMs

A language model's *vocabulary* $\mathcal{V}$ is a finite set of tokens (typically $|\mathcal{V}| \approx 5 \times 10^4$). The tokenizer is a function T : strings $\to \mathcal{V}^*$. The *embedding map* $E : \mathcal{V} \to \mathbb{R}^d$ is a function from a finite set into a real vector space; the implementation as a lookup table $E[i]$ requires that the vocabulary index $\mathcal{V} \to \{0,1,\ldots,|\mathcal{V}|-1\}$ be a *bijection.* Injectivity guarantees distinct tokens get distinct rows; surjectivity guarantees every row is reachable. We revisit this in Chapter 19, where E becomes the first learnable layer of the transformer. The rest of the book is, in a strict sense, a long story about which functions between which sets one is allowed to write down.

Chapter 2: Numbers, sequences, limits, completeness

Motivation

In Chapter 1 we built logic, sets, and proof technique. We now build the arithmetic playground in which all of analysis, optimization, and ultimately neural network training takes place: the real numbers $\mathbb{R}$. The central question of this chapter is not "what are numbers?" in a metaphysical sense but rather: *what structural property of* $\mathbb{R}$ *makes it the right home for limits?* The answer is **completeness**, and it is what will let us, many chapters later, prove that gradient descent iterates $\theta_t \in \mathbb{R}^d$ actually converge.

Definitions

We accept $\mathbb{N} = \{0,1,2,\ldots\}$ as constructed in Chapter 1 (von Neumann ordinals). The integers $\mathbb{Z}$ extend $\mathbb{N}$ by formal additive inverses; the rationals $\mathbb{Q}$ extend $\mathbb{Z}$ by formal multiplicative inverses of nonzero elements. We then take $\mathbb{R}$ to be a complete ordered field containing $\mathbb{Q}$ (constructed via Dedekind cuts or Cauchy completion of $\mathbb{Q}$). Thus

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

Definition 0.8 (Sequence)**.** *A* sequence *in a set X is a function $a : \mathbb{N} \to X$, written $(a_n)_{n\in\mathbb{N}}$.*

Definition 0.9 (Limit, ε–N)**.** *A real sequence (a_n)* converges *to $L \in \mathbb{R}$, written $a_n \to L$, iff*

$$\forall \varepsilon > 0 \ \ \exists N \in \mathbb{N} \ \ \forall n \geq N : \quad |a_n - L| < \varepsilon.$$

Definition 0.10 (Cauchy)**.** *(a_n) is* Cauchy *iff $\forall \varepsilon > 0 \, \exists N \, \forall m, n \geq N : |a_m - a_n| < \varepsilon$.*

Definition 0.11 (Bounded, monotone)**.** *(a_n) is* bounded above *iff $\exists M : a_n \leq M$ for all n; bounded below symmetrically;* bounded *iff both. It is* monotone increasing *iff $a_n \leq a_{n+1}$ for all n.*

Definition 0.12 (Supremum, infimum)**.** *Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. $u \in \mathbb{R}$ is an* upper bound *if $s \leq u$ for all $s \in S$. The* supremum $\sup S$ *is the least upper bound: an upper bound such that for every $\varepsilon > 0$ there exists $s \in S$ with $s > \sup S - \varepsilon$. The* infimum $\inf S$ *is defined dually.*

Definition 0.13 (Completeness axiom of $\mathbb{R}$)**.** *Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.*

Theorems and proofs

Theorem 0.14 (Irrationality of $\sqrt{2}$)**.** *There is no rational q with $q^2 = 2$.*

Proof. Suppose for contradiction (technique from Chapter 1) that $q = p/r$ with $p, r \in \mathbb{Z}$, $r \neq 0$, the fraction in lowest terms (so $\gcd(p, r) = 1$), and $q^2 = 2$. Then $p^2 = 2r^2$, so p^2 is even, hence p is even (since the square of an odd integer is odd). Write $p = 2k$. Substituting, $4k^2 = 2r^2$, i.e. $r^2 = 2k^2$, so r^2 is even and therefore r is even. But then $2 \mid \gcd(p, r)$, contradicting $\gcd(p, r) = 1$. □

Remark 0.15. This is the first concrete witness that $\mathbb{Q}$ is *incomplete*: the bounded-above set $S = \{q \in \mathbb{Q} : q^2 < 2\}$ has no supremum *in* $\mathbb{Q}$. The completeness axiom is exactly the patch that distinguishes $\mathbb{R}$ from $\mathbb{Q}$.

Theorem 0.16 (Bounded monotone convergence)**.** *Let (a_n) be monotone increasing and bounded above in $\mathbb{R}$. Then (a_n) converges, and $\lim a_n = \sup_n a_n$.*

Proof. Let $S = \{a_n : n \in \mathbb{N}\}$. S is nonempty (it contains a_0) and bounded above by hypothesis. By the completeness axiom, $L := \sup S$ exists in $\mathbb{R}$. Fix $\varepsilon > 0$. By the supremum's least-upper-bound property, $L - \varepsilon$ is *not* an upper bound, so there exists $N \in \mathbb{N}$ with $a_N > L - \varepsilon$. For any $n \geq N$, monotonicity gives $a_n \geq a_N > L - \varepsilon$, while $a_n \leq L$ since L is an upper bound. Hence

$$L - \varepsilon < a_n \leq L \quad \Longrightarrow \quad |a_n - L| < \varepsilon,$$

which is exactly the ε–N definition of $\lim a_n = L$. □

Lemma 0.17 (Bolzano–Weierstrass for $\mathbb{R}$)**.** *Every bounded real sequence has a convergent subsequence.*

Proof. Let $(a_n) \subseteq [c_0, d_0]$ with $c_0 < d_0$. Bisect at midpoint $m_0 = (c_0 + d_0)/2$; at least one of $[c_0, m_0]$, $[m_0, d_0]$ contains a_n for infinitely many n (else only finitely many terms total, contradicting $|\mathbb{N}| = \infty$). Call this half $[c_1, d_1]$ and pick an index n_1 with $a_{n_1} \in [c_1, d_1]$. Iterate: at stage k we have nested $[c_k, d_k]$ of length $(d_0 - c_0)/2^k$ each containing infinitely many terms, and pick $n_k > n_{k-1}$ with $a_{n_k} \in [c_k, d_k]$. The sequence (c_k) is monotone increasing and bounded above by d_0, so by Theorem 2.8 it converges to some L; similarly $(d_k) \to L$ since $d_k - c_k = (d_0 - c_0)/2^k \to 0$. Since $c_k \leq a_{n_k} \leq d_k$, the squeeze gives $a_{n_k} \to L$. □

Theorem 0.18 (Cauchy completeness of $\mathbb{R}$). *Every Cauchy sequence in $\mathbb{R}$ converges.*

Proof. Let (a_n) be Cauchy.

Step 1 (bounded). Pick $\varepsilon = 1$ and N from the Cauchy condition; for $n \geq N$, $|a_n| \leq |a_N| + 1$, so (a_n) is bounded by $M = \max(|a_0|, \ldots, |a_{N-1}|, |a_N| + 1)$.

Step 2 (convergent subsequence). By the Bolzano–Weierstrass lemma, some subsequence $a_{n_k} \to L$.

Step 3 (full sequence). Fix $\varepsilon > 0$. Choose N_1 with $|a_m - a_n| < \varepsilon/2$ for all $m, n \geq N_1$ (Cauchy property), and choose K with $n_K \geq N_1$ and $|a_{n_K} - L| < \varepsilon/2$ (subsequence convergence). Then for any $n \geq N_1$,

$$|a_n - L| \leq |a_n - a_{n_K}| + |a_{n_K} - L| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

□

Code sketch

Numerically the partial sums $S_n = \sum_{k=1}^{n} 1/k^2$ approach $\pi^2/6$. We print $|S_n - \pi^2/6|$ and locate the smallest N realizing each tolerance $\varepsilon \in \{10^{-2}, 10^{-4}, 10^{-6}\}$, contrast a Cauchy sequence $(1/n)$ with a non-Cauchy one $((-1)^n)$, run a Newton iteration to $\sqrt{2}$, and watch a bounded monotone telescoping series converge to 1.

Connection to LLMs

Training a transformer is a sequence $(\theta_t)_{t\in\mathbb{N}}$ in $\mathbb{R}^d$ produced by an optimizer (SGD, Adam). Statements like "the loss converges" or "the iterates converge to a stationary point" are *limit statements in the sense of Definition 2.2.* Convergence proofs for SGD (Chapter 13) and Adam (Chapter 14) reduce, ultimately, to bounded-monotone arguments on auxiliary scalar sequences (loss values, squared gradient norms) — and those arguments would simply be false over $\mathbb{Q}$. Completeness is not philosophical decoration; it is the load-bearing axiom under every convergence theorem in deep learning.

Chapter 3: Continuity, univariate differentiation, chain rule

Motivation

Chapter 2 built the language of limits (ε–N for sequences and ε–δ for functions). *Continuity* asserts that small input perturbations cause small output perturbations; *differentiability* strengthens this to "the perturbation is asymptotically linear." Both notions sit beneath every gradient that flows through a neural network. The chain rule, the central theorem of this chapter, is precisely the algebraic identity that backpropagation evaluates at scale (forward to Chapter 18).

Definitions

Definition 0.19 (ε–δ continuity at a point). *Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$ and $a \in D$. We say f is* continuous at *a iff*

$$\forall\, \varepsilon > 0\ \exists\, \delta > 0 :\ |x - a| < \delta \text{ and } x \in D \implies |f(x) - f(a)| < \varepsilon.$$

f is continuous on *$S \subseteq D$ iff it is continuous at every $a \in S$.*

Proposition 0.20 (Sequential characterization). *f is continuous at a iff for every sequence $x_n \to a$ in D we have $f(x_n) \to f(a)$.*

Proof. ($\Rightarrow$) Given $\varepsilon > 0$, pick δ from continuity; since $x_n \to a$, eventually $|x_n - a| < \delta$, hence $|f(x_n) - f(a)| < \varepsilon$. ($\Leftarrow$) Suppose f is not continuous at a: some $\varepsilon_0 > 0$ admits no working δ, so for each $n \in \mathbb{N}$ pick x_n with $|x_n - a| < 1/n$ but $|f(x_n) - f(a)| \geq \varepsilon_0$. Then $x_n \to a$ and yet $f(x_n) \not\to f(a)$, contradicting the hypothesis. □

Definition 0.21 (Differentiability)**.** *For a an interior point of D, f is* differentiable at a *iff*

$$f'(a) \;:=\; \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

exists in $\mathbb{R}$.

Theorems and proofs

Theorem 0.22 (Differentiable $\Rightarrow$ continuous)**.** *If f is differentiable at a, then f is continuous at a.*

Proof. For $h \neq 0$ we have the algebraic identity

$$f(a+h) - f(a) = h \cdot \frac{f(a+h) - f(a)}{h}.$$

By limit algebra (Chapter 2), the right-hand side tends to $0 \cdot f'(a) = 0$ as $h \to 0$. Hence $\lim_{h\to 0} f(a+h) = f(a)$, which is continuity at a. □

Theorem 0.23 (Sum, product, quotient rules)**.** *Let f, g be differentiable at a. Then $f+g$, fg, and (provided $g(a) \neq 0$) f/g are differentiable at a, with*

$$(f+g)'(a) = f'(a) + g'(a), \qquad (fg)'(a) = f'(a)g(a) + f(a)g'(a),$$
$$\Big(\frac{f}{g}\Big)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Proof of the product rule. Use the add-and-subtract trick:

$$\begin{aligned}\frac{(fg)(a+h) - (fg)(a)}{h} &= \frac{f(a+h)g(a+h) - f(a)g(a+h)}{h} + \frac{f(a)g(a+h) - f(a)g(a)}{h} \\ &= \frac{f(a+h) - f(a)}{h}\, g(a+h) \;+\; f(a)\, \frac{g(a+h) - g(a)}{h}.\end{aligned}$$

By Theorem 0.22, $g(a+h) \to g(a)$ as $h \to 0$. The two difference quotients tend to $f'(a)$ and $g'(a)$ respectively. Limit algebra yields $f'(a)g(a) + f(a)g'(a)$. Sum and quotient rules are analogous (use the algebraic identity $\frac{1}{g(a+h)} - \frac{1}{g(a)} = -\frac{g(a+h)-g(a)}{g(a+h)g(a)}$ and continuity of g). □

Theorem 0.24 (Chain rule, Carathéodory form)**.** *Let g be differentiable at a and f differentiable at $b := g(a)$. Then $f \circ g$ is differentiable at a and*

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

Proof. Define the *Carathéodory function*

$$\phi(y) := \begin{cases} \dfrac{f(y) - f(b)}{y - b}, & y \neq b, \\ f'(b), & y = b. \end{cases}$$

By the definition of $f'(b)$, ϕ is continuous at b. For *every* y in a neighborhood of b,

$$f(y) - f(b) = \phi(y)\,(y - b), \tag{$\star$}$$

which holds at $y = b$ trivially and by construction otherwise — a key advantage: no division by zero is ever required. Apply ($\star$) with $y = g(x)$:

$$\frac{f(g(x)) - f(g(a))}{x - a} = \phi(g(x)) \cdot \frac{g(x) - g(a)}{x - a}.$$

As $x \to a$, $g(x) \to g(a) = b$ by Theorem 0.22, and ϕ is continuous at b, so $\phi(g(x)) \to \phi(b) = f'(b)$ by Proposition 0.20. The second factor tends to $g'(a)$. Limit algebra closes the proof. □

Remark 0.25. The Carathéodory device sidesteps the classical pitfall of the difference-quotient proof, where one must handle the case $g(x) = g(a)$ for x arbitrarily close to a (e.g. g constant on a sequence). Here ϕ is defined everywhere, so the identity ($\star$) requires no caveat.

Lemma 0.26 (Rolle)**.** *If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Proof. By the extreme value theorem (Chapter 4, via compactness of $[a, b]$), f attains its maximum M and minimum m on $[a, b]$. If $M = m$ then f is constant and any $c \in (a, b)$ works. Otherwise an extremum is attained at some interior $c \in (a, b)$; Fermat's interior-extremum lemma — one-sided difference quotients have opposite signs at an interior extremum, so the two-sided limit $f'(c)$ must equal 0 — finishes the job. □

Theorem 0.27 (Mean value theorem)**.** *If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define the auxiliary function

$$h(x) := f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then h is continuous on $[a, b]$ (sum/product of continuous functions) and differentiable on (a, b) with

$$h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Direct computation gives $h(a) = f(a)$ and $h(b) = f(b) - (f(b) - f(a)) = f(a)$, so $h(a) = h(b)$. By Lemma 0.26 there exists $c \in (a, b)$ with $h'(c) = 0$, i.e. $f'(c) = (f(b) - f(a))/(b - a)$. □

Code sketch

The companion notebook (`cells.json`) (i) certifies ε–δ continuity for $f(x) = x^2$ at $a = 2$ by binary-search on δ for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$; (ii) plots the $O(h)$ error of the forward-difference approximation to $\sin'$; (iii) verifies the chain rule for $f(x) = \sin(x^2)$ at $x = 1$ to $\sim 10^{-7}$; (iv) finds an MVT witness $c \in (0, 2)$ for $f(x) = x^3 - 2x$ by bisection.

Connection to LLMs

A transformer is a composition $L_K \circ L_{K-1} \circ \cdots \circ L_1$ of differentiable layers. The gradient of the loss with respect to any parameter θ in layer i is, by iterated application of Theorem 0.24,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial L_K} \cdot \frac{\partial L_K}{\partial L_{K-1}} \cdots \frac{\partial L_{i+1}}{\partial L_i} \cdot \frac{\partial L_i}{\partial \theta}.$$

Backpropagation is precisely the right-to-left evaluation of this product (Chapter 18). The MVT, in turn, underwrites convergence proofs for gradient descent (Chapter 14): it bounds $|f(x) - f(y)| \leq \sup |f'| \cdot |x - y|$, the Lipschitz hypothesis under which descent provably decreases the loss. Without the chain rule, deep learning does not exist.

Chapter 4: Multivariate calculus: partials, gradients, Jacobians

Motivation

A neural network is a function $F : \mathbb{R}^n \to \mathbb{R}^m$ built by composing many simple maps; training it requires the gradient of a scalar loss with respect to a high-dimensional parameter vector. Chapter 3 handled the one-dimensional theory: existence of $f'(a)$, the mean value theorem (MVT), and the single-variable chain rule through the Carathéodory factorization $f(x) - f(a) = \varphi(x)(x - a)$ with φ continuous at a. We now lift these ideas to several variables. Two pitfalls: (i) the *correct* notion of differentiability is *not* "all partials exist" but the stronger Fréchet condition, and (ii) the chain rule becomes a matrix product—the operation iterated by backpropagation in Chapter 18.

Definitions

Let $f : U \to \mathbb{R}^m$ with $U \subseteq \mathbb{R}^n$ open, $\mathbf{a} \in U$, and $\mathbf{e}_i$ the i-th standard basis vector.

Definition 0.28 (Partial derivative)**.** $\dfrac{\partial f}{\partial x_i}(\mathbf{a}) := \lim_{h \to 0} \dfrac{f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a})}{h}$*, when the limit exists.*

Definition 0.29 (Directional derivative)**.** *For a unit vector* $\mathbf{v}$*,* $D_{\mathbf{v}} f(\mathbf{a}) := \lim_{t \to 0} \dfrac{f(\mathbf{a} + t\mathbf{v}) - f(\mathbf{a})}{t}$.

Definition 0.30 (Fréchet differentiability)**.** f *is* differentiable *at* $\mathbf{a}$ *if there exists a linear* $L : \mathbb{R}^n \to \mathbb{R}^m$ *with*

$$\lim_{\mathbf{h} \to \mathbf{0}} \frac{\|f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - L\mathbf{h}\|}{\|\mathbf{h}\|} = 0,$$

i.e. the remainder is $o(\|\mathbf{h}\|)$*. The matrix of* L *in the standard basis is the* Jacobian $J_f(\mathbf{a}) \in \mathbb{R}^{m \times n}$*. When* $m = 1$*, the* gradient *is* $\nabla f(\mathbf{a}) := J_f(\mathbf{a})^\top$*.*

Remark 0.31. Differentiability is strictly stronger than existence of partials: $f(x, y) = xy/(x^2 + y^2)$ (with $f(0, 0) = 0$) has both partials zero at the origin, yet f is discontinuous there and so not Fréchet differentiable.

Theorems and proofs

Theorem 0.32 (Differentiable implies partials exist; $L = J_f$)**.** *If* f *is differentiable at* $\mathbf{a}$ *with derivative* L*, then* $\partial f/\partial x_j(\mathbf{a})$ *exists for every* j *and equals the* j*-th column of the matrix of* L*.*

Proof. Take $\mathbf{h} = h\mathbf{e}_j$, $h \neq 0$. Differentiability yields

$$f(\mathbf{a} + h\mathbf{e}_j) - f(\mathbf{a}) = h\, L\mathbf{e}_j + r(h), \qquad \|r(h)\|/|h| \to 0.$$

Divide by h and let $h \to 0$: the left side tends to $\partial f/\partial x_j(\mathbf{a})$ by Definition 1, the right side to $L\mathbf{e}_j$. Hence the partial exists and equals the j-th column of L. □

Theorem 0.33 (Continuous partials imply differentiability)**.** *If all $\partial f/\partial x_j$ exist on a neighborhood of $\mathbf{a}$ and are continuous at $\mathbf{a}$, then f is differentiable at $\mathbf{a}$.*

Sketch. Reduce to scalar codomain componentwise. Telescope along axes:

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = \sum_{j=1}^{n} \Big[f(\mathbf{a} + \mathbf{h}_{<j} + h_j\mathbf{e}_j) - f(\mathbf{a} + \mathbf{h}_{<j}) \Big], \qquad \mathbf{h}_{<j} := \sum_{i<j} h_i\mathbf{e}_i.$$

The single-variable MVT (Chapter 3) applied to the j-th summand produces ξ_j between 0 and h_j with

$$f(\mathbf{a} + \mathbf{h}_{<j} + h_j\mathbf{e}_j) - f(\mathbf{a} + \mathbf{h}_{<j}) = h_j\, \partial_j f(\mathbf{a} + \mathbf{h}_{<j} + \xi_j\mathbf{e}_j).$$

Subtract $\sum_j h_j\, \partial_j f(\mathbf{a})$. By continuity of each $\partial_j f$, the difference $|\partial_j f(\cdot) - \partial_j f(\mathbf{a})| \to 0$ as $\mathbf{h} \to \mathbf{0}$. Cauchy–Schwarz then bounds the remainder by $\|\mathbf{h}\| \cdot \varepsilon(\mathbf{h})$ with $\varepsilon \to 0$, i.e. $o(\|\mathbf{h}\|)$. (Spivak, *Calculus on Manifolds*, Thm. 2-8.) □

Theorem 0.34 (Multivariate chain rule)**.** *If $g : \mathbb{R}^k \to \mathbb{R}^n$ is differentiable at $\mathbf{a}$ and $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{b} = g(\mathbf{a})$, then $f \circ g$ is differentiable at $\mathbf{a}$ with*

$$J_{f\circ g}(\mathbf{a}) = J_f(\mathbf{b})\, J_g(\mathbf{a}).$$

Proof. Write the Carathéodory-style remainders $\rho_g(\mathbf{h}) := g(\mathbf{a}+\mathbf{h}) - g(\mathbf{a}) - J_g(\mathbf{a})\mathbf{h} = o(\|\mathbf{h}\|)$, and $\rho_f(\mathbf{k}) = o(\|\mathbf{k}\|)$. Let $\mathbf{k}(\mathbf{h}) := g(\mathbf{a}+\mathbf{h}) - g(\mathbf{a})$. Then

$$\begin{aligned} f(g(\mathbf{a}+\mathbf{h})) - f(g(\mathbf{a})) &= J_f(\mathbf{b})\, \mathbf{k}(\mathbf{h}) + \rho_f(\mathbf{k}(\mathbf{h})) \\ &= J_f(\mathbf{b})J_g(\mathbf{a})\mathbf{h} + J_f(\mathbf{b})\rho_g(\mathbf{h}) + \rho_f(\mathbf{k}(\mathbf{h})). \end{aligned}$$

Bound: $\|J_f(\mathbf{b})\rho_g(\mathbf{h})\| \le \|J_f(\mathbf{b})\|_{\mathrm{op}} \cdot o(\|\mathbf{h}\|) = o(\|\mathbf{h}\|)$. For the last term, $\|\mathbf{k}(\mathbf{h})\| \le (\|J_g(\mathbf{a})\|_{\mathrm{op}} + 1)\|\mathbf{h}\|$ for small $\|\mathbf{h}\|$, hence $\rho_f(\mathbf{k}(\mathbf{h})) = o(\|\mathbf{k}\|) = o(\|\mathbf{h}\|)$. The total remainder is $o(\|\mathbf{h}\|)$, so $f \circ g$ is differentiable with derivative $J_f(\mathbf{b})J_g(\mathbf{a})$. □

Theorem 0.35 (Schwarz / Clairaut)**.** *If $f : U \to \mathbb{R}$ is C^2 on U open and $\mathbf{a} \in U$, then for all i, j*

$$\frac{\partial^2 f}{\partial x_i\, \partial x_j}(\mathbf{a}) = \frac{\partial^2 f}{\partial x_j\, \partial x_i}(\mathbf{a}).$$

Proof. Fix $i \neq j$ and restrict to the two-variable slice in coordinates x_i, x_j (other coordinates frozen at the values of $\mathbf{a}$). Define the second difference

$$\Delta(h,k) := f(\mathbf{a} + h\mathbf{e}_i + k\mathbf{e}_j) - f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a} + k\mathbf{e}_j) + f(\mathbf{a}).$$

Let $\varphi(s) := f(\mathbf{a} + s\mathbf{e}_i + k\mathbf{e}_j) - f(\mathbf{a} + s\mathbf{e}_i)$, so $\Delta(h,k) = \varphi(h) - \varphi(0)$. By MVT in s: $\Delta = h\,\varphi'(\xi)$ for some $\xi \in (0,h)$, with

$$\varphi'(\xi) = \partial_i f(\mathbf{a} + \xi\mathbf{e}_i + k\mathbf{e}_j) - \partial_i f(\mathbf{a} + \xi\mathbf{e}_i).$$

Apply MVT again, this time in k: $\varphi'(\xi) = k\, \partial_j\partial_i f(\mathbf{a} + \xi\mathbf{e}_i + \eta\mathbf{e}_j)$ for some $\eta \in (0,k)$. Hence

$$\Delta(h,k) = hk\, \partial_j\partial_i f(\mathbf{a} + \xi\mathbf{e}_i + \eta\mathbf{e}_j).$$

By the symmetric argument starting with $\psi(t) := f(\mathbf{a}+h\mathbf{e}_i+t\mathbf{e}_j) - f(\mathbf{a}+t\mathbf{e}_j)$, $\Delta(h,k) = hk\, \partial_i\partial_j f(\mathbf{a} + \xi'\mathbf{e}_i + \eta'\mathbf{e}_j)$. Divide by hk and let $(h,k) \to (0,0)$. Continuity of the second partials (C^2) makes both sides converge to $\partial_j\partial_i f(\mathbf{a})$ and $\partial_i\partial_j f(\mathbf{a})$ respectively, which are therefore equal. □

Code sketch

The accompanying notebook verifies every theorem numerically: gradient of $f(x, y) = x^2y+\sin(x+y)$ via central differences; Jacobian of $\mathbf{f}(x, y, z) = (xyz,\, x^2 + \sin y + e^z)$ column-by-column; chain rule on $f \circ g$ with $g(t) = (\cos t, \sin t),\; f(x, y) = x^2 + y^2$ (composite identically 1, so the chain-rule product must vanish); equality of mixed partials for $f(x, y) = x^3y^2 + \sin(xy)$.

Connection to LLMs

A transformer is a composition $F = F_L \circ \cdots \circ F_1$ of differentiable layers, scored by a scalar loss $\mathcal{L} = \ell \circ F$. Theorem 4.7 states

$$\nabla_{\theta_\ell}\mathcal{L} = \big(J_\ell(F(x))\, J_{F_L} \cdots J_{F_{\ell+1}}\big)\, \frac{\partial F_\ell}{\partial \theta_\ell}.$$

Backpropagation never materializes any J_{F_k}. It propagates the row vector $\mathbf{v}^\top := J_\ell\, J_{F_L} \cdots J_{F_{\ell+1}}$ right-to-left via *vector–Jacobian products* (VJPs)—one per layer, each at O(forward cost). Reverse-mode autodiff is precisely Theorem 4.7 with associativity exploited; we derive the algorithm formally in Chapter 18.

Chapter 5: Linear algebra I: vector spaces, basis, linear maps

Motivation

Every transformer layer is, between its bias terms and nonlinearities, a *linear map between finite-dimensional real vector spaces.* The "hidden dimension" d is just $\dim \mathbb{R}^d$; a "weight matrix" $W \in \mathbb{R}^{m\times n}$ is the matrix representation of a linear map $\mathbb{R}^n \to \mathbb{R}^m$; "residual connections" are addition in a vector space; "low-rank adapters" are statements about the image of a linear map. Before attention (Chapter 21) or embeddings (Chapter 19), we need the grammar of vector spaces, bases, dimension, kernels, images, and the rank–nullity theorem. We use Chapter 1 (sets, functions, logic) throughout.

Definitions

Definition 0.36 (Field)**.** *A* field $\mathbb{F}$ *is a set with operations* $+, \cdot$ *such that* $(\mathbb{F}, +)$ *is an abelian group with identity* 0*,* $(\mathbb{F} \setminus \{0\}, \cdot)$ *is an abelian group with identity* 1*, and multiplication distributes over addition. The canonical example is* $\mathbb{F} = \mathbb{R}$*.*

Definition 0.37 (Vector space, eight axioms)**.** *A* vector space V *over* $\mathbb{F}$ *is a set with addition* $+\colon V \times V \to V$ *and scalar multiplication* $\cdot\colon \mathbb{F} \times V \to V$ *satisfying, for all* $u, v, w \in V$ *and* $a, b \in \mathbb{F}$*:*

1. $(u + v) + w = u + (v + w)$*;*
2. $u + v = v + u$*;*
3. $\exists 0 \in V$ *with* $v + 0 = v$ *for all* v*;*
4. $\forall v\, \exists(-v)$ *with* $v + (-v) = 0$*;*
5. $a \cdot (u + v) = a \cdot u + a \cdot v$*;*
6. $(a + b) \cdot v = a \cdot v + b \cdot v$*;*

7. $(ab) \cdot v = a \cdot (b \cdot v)$*;*

8. $1 \cdot v = v$*.*

Definition 0.38 (Subspace, span, independence, basis, dimension)**.** *A* subspace $U \subset V$ *is a subset closed under* $+$ *and scalar multiplication that contains* 0*. A* linear combination *of* $v_1, \ldots, v_k$ *is* $\sum a_i v_i$ *with* $a_i \in \mathbb{F}$*. The* span *of* $S \subset V$ *is* $\operatorname{span}(S) := \{\sum a_i v_i : v_i \in S, a_i \in \mathbb{F}\}$*, the smallest subspace containing* S*. A finite set* $\{v_1, \ldots, v_k\}$ *is* linearly independent *iff* $\sum a_i v_i = 0 \Rightarrow a_1 = \cdots = a_k = 0$*. A* basis *of* V *is a linearly independent spanning set; the* dimension $\dim V$ *is its cardinality (well-defined by Theorem 0.192 below).*

Definition 0.39 (Linear map, kernel, image, matrix representation)**.** *A map* $T\colon V \to W$ *between vector spaces over the same* $\mathbb{F}$ *is* linear *iff* $T(au + bv) = aT(u) + bT(v)$*. Define* $\ker T := \{v \in V : Tv = 0\}$ *and* $\operatorname{im} T := \{Tv : v \in V\}$*; both are subspaces. Given bases* (e_j) *of* $\mathbb{R}^n$ *and* (f_i) *of* $\mathbb{R}^m$*, the* matrix representation *of* T *is* $A \in \mathbb{R}^{m\times n}$ *with* $T(e_j) = \sum_i A_{ij} f_i$*.*

Theorems and proofs

Lemma 0.40 (Steinitz exchange)**.** *Let* V *be a vector space over* $\mathbb{F}$*. If* $\{v_1, \ldots, v_m\}$ *is linearly independent and* $\{w_1, \ldots, w_n\}$ *spans* V*, then* $m \le n$*, and (after reindexing the* w_j*) we may replace* m *of them by* $v_1, \ldots, v_m$ *so that the resulting set still spans* V*.*

Proof. By induction on m. The case $m = 0$ is trivial. Assume the claim for $m-1$: after reindexing, $\{v_1, \ldots, v_{m-1}, w_m, \ldots, w_n\}$ spans V. In particular $m - 1 \le n$. Then

$$v_m = \sum_{i<m} a_i v_i + \sum_{j\ge m} b_j w_j$$

for some scalars. If all $b_j = 0$, then $v_m \in \operatorname{span}(v_1, \ldots, v_{m-1})$, contradicting linear independence of $\{v_1, \ldots, v_m\}$. Hence some $b_{j_0} \ne 0$ for $j_0 \in \{m, \ldots, n\}$, forcing $n \ge m$. Reindex so $j_0 = m$ and solve:

$$w_m = b_m^{-1}\Big(v_m - \sum_{i<m} a_i v_i - \sum_{j>m} b_j w_j\Big) \in \operatorname{span}(v_1, \ldots, v_m, w_{m+1}, \ldots, w_n).$$

Therefore $\operatorname{span}(v_1, \ldots, v_m, w_{m+1}, \ldots, w_n)$ contains $\{v_1, \ldots, v_{m-1}, w_m, \ldots, w_n\}$, which spans V, so the new set spans V. □

Theorem 0.41 (Spanning sets contain bases; independent sets extend to bases)**.** *In a finite-dimensional* V*: (a) any finite spanning set contains a basis; (b) any linearly independent set extends to a basis.*

Proof. (a) If a spanning set S is dependent, some $w \in S$ lies in $\operatorname{span}(S \setminus \{w\})$, so $S \setminus \{w\}$ still spans. Iterate; finiteness terminates the process at an independent spanning set.
(b) Given an independent set L and a finite spanning set S, apply Lemma 0.40 to replace $|L|$ of the w's by L, producing a spanning set containing L. Then apply (a), removing only S-side elements. □

Theorem 0.42 (Invariance of dimension)**.** *Any two bases of a finite-dimensional* V *have the same cardinality.*

Proof. Let $\mathcal{B}_1, \mathcal{B}_2$ be bases with $|\mathcal{B}_1| = m$, $|\mathcal{B}_2| = n$. Since $\mathcal{B}_1$ is independent and $\mathcal{B}_2$ spans, Lemma 0.40 gives $m \le n$. Swap roles: $n \le m$. Hence $m = n$. □

Theorem 0.43 (Rank–nullity)**.** *Let $T \colon V \to W$ be linear with $\dim V = n < \infty$. Then*

$$\dim\ker T + \dim\operatorname{im} T \;=\; n.$$

Proof. Let $(u_1, \ldots, u_k)$ be a basis of $\ker T$. By Theorem 0.41(b), extend to a basis $(u_1, \ldots, u_k, v_1, \ldots, v_{n-k})$ of V. We claim $(Tv_1, \ldots, Tv_{n-k})$ is a basis of $\operatorname{im} T$.

Spanning. Any $w \in \operatorname{im} T$ has $w = T(\sum a_i u_i + \sum b_j v_j) = \sum b_j T v_j$ since $Tu_i = 0$.

Independence. If $\sum c_j T v_j = 0$, then $T(\sum c_j v_j) = 0$, so $\sum c_j v_j \in \ker T = \operatorname{span}(u_i)$. Write $\sum c_j v_j = \sum d_i u_i$, i.e. $\sum c_j v_j - \sum d_i u_i = 0$. Independence of the full basis forces all $c_j = 0$.

Hence $\dim\operatorname{im} T = n - k = n - \dim\ker T$. □

Code sketch

We will (a) implement `is_linearly_independent` and `dim_span` via `numpy.linalg.matrix_rank`; (b) build a 4×6 random integer matrix, compute rank and a basis of $\ker A$ via the right singular vectors of A corresponding to zero singular values, verifying $Av = 0$; (c) verify the change-of-basis formula $Av = P(P^{-1}AP)(P^{-1}v)$ for a random invertible P; (d) confirm $\operatorname{rank}(A) + \dim\ker A = 6$.

Connection to LLMs

A transformer with hidden dimension d operates on the vector space $\mathbb{R}^d$. The query/key/value projections in attention and the up/down projections in the MLP are linear maps $\mathbb{R}^d \to \mathbb{R}^{d_k}$ or $\mathbb{R}^d \to \mathbb{R}^{d_{\mathrm{ff}}}$ (Chapter 21). The token embedding (Chapter 19) is a linear map $\mathbb{R}^{|\mathcal{V}|} \to \mathbb{R}^d$ applied to a one-hot input, i.e. a row lookup. The "rank" of an attention matrix and the "intrinsic dimension" of activations are claims about $\dim\operatorname{im}$. LoRA fine-tuning constrains weight updates to a low-dimensional subspace — a direct use of $\dim\operatorname{im} T \leq \min(m, n)$. Rank–nullity returns when we count free parameters and degrees of freedom.

Chapter 6: Linear algebra II: inner products, norms, eigenvalues, SVD

Motivation

Chapter 5 built vector spaces and linear maps as raw algebraic objects. To do *geometry* — length, angle, orthogonality, best approximation — we add a single piece of structure: an *inner product.* From it we will derive the Cauchy–Schwarz inequality, the triangle inequality, the spectral theorem for symmetric matrices, the singular value decomposition (SVD), and the Eckart–Young low-rank approximation theorem. These are the load-bearing tools behind PCA, attention, and LoRA fine-tuning of LLMs.

Definitions

Definition 0.44 (Inner product)**.** *A function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ on a real vector space V is an* inner product *if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $a, b \in \mathbb{R}$:*

1. *Symmetry: $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$.*
2. *Linearity in the first argument: $\langle a\mathbf{x} + b\mathbf{y}, \mathbf{z} \rangle = a\langle \mathbf{x}, \mathbf{z} \rangle + b\langle \mathbf{y}, \mathbf{z} \rangle$.*
3. *Positive-definiteness: $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ with equality iff $\mathbf{x} = \mathbf{0}$.*

The canonical example on $\mathbb{R}^n$ is the dot product $\langle \mathbf{x}, \mathbf{y}\rangle = \sum_i x_i y_i = \mathbf{x}^\top \mathbf{y}$.

Definition 0.45 (Induced norm)**.** $\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle}$.

Definition 0.46 (Orthogonality)**.** $\mathbf{x}, \mathbf{y}$ *are* orthogonal *if* $\langle \mathbf{x}, \mathbf{y}\rangle = 0$*. A set* $\{\mathbf{q}_i\}$ *is* orthonormal *if* $\langle \mathbf{q}_i, \mathbf{q}_j\rangle = \delta_{ij}$*. An* orthonormal basis *is an orthonormal spanning set.*

Definition 0.47 (Eigenvalue, eigenvector, characteristic polynomial)**.** *For* $A \in \mathbb{R}^{n\times n}$*,* $(\lambda, \mathbf{v})$ *with* $\mathbf{v} \neq \mathbf{0}$ *is an* eigenpair *if* $A\mathbf{v} = \lambda\mathbf{v}$*. The* characteristic polynomial *is* $p_A(\lambda) := \det(A - \lambda I)$*; its roots are the eigenvalues.*

Definition 0.48 (Symmetric, positive (semi)definite)**.** *A is* symmetric *if* $A^\top = A$*;* positive semidefinite *(PSD) if symmetric with* $\mathbf{x}^\top A\mathbf{x} \geq 0$ *for all* $\mathbf{x}$*;* positive definite *if strict for* $\mathbf{x} \neq \mathbf{0}$*.*

Definition 0.49 (Singular value decomposition)**.** *A* singular value decomposition *of* $A \in \mathbb{R}^{m\times n}$ *is a factorization* $A = U\Sigma V^\top$ *where* $U \in \mathbb{R}^{m\times m}$*,* $V \in \mathbb{R}^{n\times n}$ *are orthogonal* ($U^\top U = I$*,* $V^\top V = I$) *and* $\Sigma \in \mathbb{R}^{m\times n}$ *is "diagonal" with nonnegative entries* $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$*.*

Theorems and Proofs

Theorem 0.50 (Cauchy–Schwarz)**.** *For all* $\mathbf{x}, \mathbf{y} \in V$*,* $|\langle \mathbf{x}, \mathbf{y}\rangle| \leq \|\mathbf{x}\|\,\|\mathbf{y}\|$*.*

Proof. If $\mathbf{y} = \mathbf{0}$, both sides vanish. Otherwise, for every $t \in \mathbb{R}$,

$$0 \leq \|\mathbf{x} - t\mathbf{y}\|^2 = \langle \mathbf{x} - t\mathbf{y}, \mathbf{x} - t\mathbf{y}\rangle = \|\mathbf{x}\|^2 - 2t\langle \mathbf{x}, \mathbf{y}\rangle + t^2\|\mathbf{y}\|^2.$$

This is a quadratic in t that is nonnegative everywhere, so its discriminant satisfies

$$(2\langle \mathbf{x}, \mathbf{y}\rangle)^2 - 4\|\mathbf{x}\|^2\|\mathbf{y}\|^2 \leq 0,$$

i.e. $\langle \mathbf{x}, \mathbf{y}\rangle^2 \leq \|\mathbf{x}\|^2\|\mathbf{y}\|^2$. Taking square roots completes the proof. □

Corollary 0.51 (Triangle inequality)**.** $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

Proof. Expand and apply Cauchy–Schwarz:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y}\rangle + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2.$$ □

Theorem 0.52 (Spectral theorem, real symmetric case)**.** *Every symmetric* $A \in \mathbb{R}^{n\times n}$ *admits a factorization* $A = Q\Lambda Q^\top$ *with Q orthogonal and Λ real diagonal.*

Proof. Induct on n. For $n = 1$, trivial. Assume the result for $n - 1$. The Rayleigh quotient $R(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ is continuous on the unit sphere $S^{n-1} = \{\mathbf{x} : \|\mathbf{x}\| = 1\}$, which is compact, so R attains a maximum at some $\mathbf{v}_1 \in S^{n-1}$ with value λ_1. By Lagrange multipliers (or by directly differentiating $R(\mathbf{v}_1 + t\mathbf{w})$ along any tangent $\mathbf{w} \perp \mathbf{v}_1$ at $t = 0$), $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$. The eigenvalue is real because $\lambda_1 = \mathbf{v}_1^\top A\mathbf{v}_1 \in \mathbb{R}$ and $\mathbf{v}_1 \in \mathbb{R}^n$. (Alternatively, if $A\mathbf{z} = \mu\mathbf{z}$ over $\mathbb{C}$ with $\mathbf{z} \neq \mathbf{0}$, then $\mu\,\overline{\mathbf{z}}^\top\mathbf{z} = \overline{\mathbf{z}}^\top A\mathbf{z} = (A\overline{\mathbf{z}})^\top\mathbf{z} = \overline{\mu}\,\overline{\mathbf{z}}^\top\mathbf{z}$ since A is real symmetric, so $\mu = \overline{\mu}$.)

Let $W = \{\mathbf{v}_1\}^\perp$, an $(n-1)$-dimensional subspace. For $\mathbf{w} \in W$,

$$\langle A\mathbf{w}, \mathbf{v}_1\rangle = \langle \mathbf{w}, A\mathbf{v}_1\rangle = \lambda_1\langle \mathbf{w}, \mathbf{v}_1\rangle = 0,$$

so A maps $W \to W$. The restriction $A|_W$ is symmetric with respect to the inherited inner product. By induction there is an orthonormal eigenbasis $\mathbf{v}_2, \ldots, \mathbf{v}_n$ of W with eigenvalues $\lambda_2, \ldots, \lambda_n$. Then $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is an orthonormal eigenbasis of A. Setting $Q = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$ and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ gives $A = Q\Lambda Q^\top$. □

Theorem 0.53 (Existence of SVD)**.** *Every $A \in \mathbb{R}^{m\times n}$ admits an SVD $A = U\Sigma V^\top$.*

Sketch. The matrix $A^\top A \in \mathbb{R}^{n\times n}$ is symmetric and PSD, since $\mathbf{x}^\top A^\top A\mathbf{x} = \|A\mathbf{x}\|^2 \geq 0$. By the spectral theorem $A^\top A = V\Lambda V^\top$ with V orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, $\lambda_i \geq 0$, sorted decreasingly. Set $\sigma_i := \sqrt{\lambda_i}$. For each i with $\sigma_i > 0$ define $\mathbf{u}_i := A\mathbf{v}_i/\sigma_i \in \mathbb{R}^m$; then

$$\langle \mathbf{u}_i, \mathbf{u}_j\rangle = \frac{\mathbf{v}_i^\top A^\top A\mathbf{v}_j}{\sigma_i\sigma_j} = \frac{\lambda_j\delta_{ij}}{\sigma_i\sigma_j} = \delta_{ij},$$

so the $\mathbf{u}_i$ are orthonormal in $\mathbb{R}^m$. Extend to an orthonormal basis to form U. By construction $AV = U\Sigma$, hence $A = U\Sigma V^\top$. □

Theorem 0.54 (Eckart–Young)**.** *Let $A = U\Sigma V^\top$ have singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$. The truncated SVD $A_k := \sum_{i=1}^k \sigma_i\mathbf{u}_i\mathbf{v}_i^\top$ minimises $\|A - B\|_F$ (and $\|A - B\|_2$) over rank-k matrices B, with $\|A - A_k\|_F^2 = \sum_{i>k}\sigma_i^2$.*

Sketch. The Frobenius norm is unitarily invariant: $\|A - B\|_F = \|U^\top(A - B)V\|_F$. The problem reduces to: among rank-k matrices $C \in \mathbb{R}^{m\times n}$, minimise

$$\|\Sigma - C\|_F^2 = \sum_i(\sigma_i - c_{ii})^2 + \sum_{i\neq j} c_{ij}^2.$$

Off-diagonal entries should be zero, and the diagonal of C should equal σ_i for $i \leq k$ and 0 for $i > k$ (matching the largest gaps), recovering A_k. The operator-norm version uses the Courant–Fischer minimax characterisation of singular values. □

Code Sketch

The accompanying notebook (`cells.json`) (i) verifies Cauchy–Schwarz on 50 random pairs in $\mathbb{R}^{10}$, (ii) computes a symmetric eigendecomposition via `numpy.linalg.eigh` and checks $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ plus orthonormality, (iii) computes an SVD via `numpy.linalg.svd` and reconstructs A, and (iv) shows that Frobenius errors of rank-1 and rank-2 approximations decrease monotonically, consistent with Eckart–Young.

Connection to LLMs

Inner products and SVD are the geometric backbone of transformers.

- **Attention** (Chapters 21–22). The score matrix $QK^\top$ is the matrix of pairwise inner products of query and key embeddings. Linear-attention and Performer-style methods exploit the empirical low-rank-ness of $QK^\top$, which by Eckart–Young is best captured by truncated SVD.
- **PCA / interpretability.** SVD of an embedding matrix yields principal components; leading singular vectors expose semantic axes used in probing studies.
- **LoRA fine-tuning.** The update $W_0 + BA$ with $B \in \mathbb{R}^{m\times r}$, $A \in \mathbb{R}^{r\times n}$, $r \ll \min(m, n)$, is literally a rank-r correction; Eckart–Young guarantees its optimality at that rank in Frobenius norm.
- **Spectral norm regularisation.** Bounding $\sigma_1(W)$ controls Lipschitz constants and stabilises training.

Every later geometric statement in this book — projections, least squares, gradient flow on weight matrices, contractive maps — descends from the inequalities and decompositions proven in this chapter.

Chapter 7: Convexity and optimization; gradient descent convergence

Motivation

Chapters 3 and 4 built derivatives, the chain rule, and gradients; Chapter 6 equipped $\mathbb{R}^n$ with a norm and the Cauchy–Schwarz inequality. With those in hand we can finally ask the optimization question that drives every line of training code in this book: given $f : \mathbb{R}^n \to \mathbb{R}$, what does the iteration $x_{t+1} = x_t - \eta \nabla f(x_t)$ actually do, and when does it converge?

The honest answer for a transformer's loss is "we don't know": it is non-convex, the iterates are stochastic (Chapter 13), and rigorous global convergence is open. But the *convex* case admits complete proofs, and those proofs supply the only quantitative vocabulary we have for the non-convex one — smoothness, condition number, descent, contraction. Phrases like "smooth loss landscape," "warmup," and "$1/L$ step size" trace directly back to the two theorems below.

Definitions

Definition 0.55 (Convex set). *$C \subseteq \mathbb{R}^n$ is* convex *iff for all $x, y \in C$ and $t \in [0, 1]$, $tx+(1-t)y \in C$.*

Definition 0.56 (Convex function). *$f : \mathbb{R}^n \to \mathbb{R}$ is* convex *iff*

$$f(tx + (1 - t)y) \le tf(x) + (1 - t)f(y) \quad \forall x, y \in \mathbb{R}^n,\ t \in [0, 1].$$

When f is differentiable this is equivalent to the first-order *characterization $f(y) \ge f(x)+\langle \nabla f(x), y-x\rangle$: every tangent hyperplane lies below the graph.*

Definition 0.57 (L-smoothness). *f is* L-smooth *iff ∇f is L-Lipschitz: $\|\nabla f(x)-\nabla f(y)\| \le L\|x-y\|$ for all x, y, with the Euclidean norm of Chapter 6.*

Definition 0.58 (μ-strong convexity). *f is* μ-strongly convex *iff for all x, y,*

$$f(y) \ge f(x) + \langle \nabla f(x), y - x\rangle + \tfrac{\mu}{2}\|y - x\|^2.$$

The ratio $\kappa = L/\mu \ge 1$ is the condition number.

Definition 0.59 (Gradient descent). *Fix $x_0 \in \mathbb{R}^n$, step size $\eta > 0$. The* gradient descent *iterates are $x_{t+1} = x_t - \eta \nabla f(x_t)$ for $t = 0, 1, 2, \ldots$.*

Theorems and proofs

Lemma 0.60 (Descent lemma). *If f is L-smooth, then for all $x, y \in \mathbb{R}^n$,*

$$f(y) \le f(x) + \langle \nabla f(x), y - x\rangle + \tfrac{L}{2}\|y - x\|^2.$$

Proof. Let $g(t) = f(x + t(y - x))$ on $[0, 1]$. By the chain rule (Chapter 4), $g'(t) = \langle \nabla f(x + t(y - x)), y - x\rangle$, and the fundamental theorem of calculus gives

$$f(y) - f(x) = g(1) - g(0) = \int_0^1 \langle \nabla f(x + t(y - x)),\, y - x\rangle\, dt.$$

Subtract $\langle \nabla f(x), y - x\rangle$ from both sides:

$$f(y) - f(x) - \langle \nabla f(x), y - x\rangle = \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x),\, y - x\rangle\, dt.$$

By Cauchy–Schwarz (Chapter 6) and the Lipschitz hypothesis,

$$\langle \nabla f(x+t(y-x)) - \nabla f(x),\, y-x\rangle \le \|\nabla f(x+t(y-x)) - \nabla f(x)\|\,\|y-x\| \le Lt\|y-x\|^2.$$

Integrating, $\int_0^1 Lt\|y-x\|^2\,dt = \frac{L}{2}\|y-x\|^2$. □

Theorem 0.61 (GD on L-smooth convex)**.** *Let f be convex and L-smooth with minimizer x^* and minimum $f^* = f(x^*)$. Then gradient descent with $\eta = 1/L$ satisfies*

$$f(x_T) - f^* \;\le\; \frac{L\|x_0 - x^*\|^2}{2T}, \qquad T \ge 1.$$

Proof. Apply the descent lemma at $y = x_{t+1} = x_t - \frac{1}{L}\nabla f(x_t)$, $x = x_t$:

$$f(x_{t+1}) \le f(x_t) - \tfrac{1}{2L}\|\nabla f(x_t)\|^2. \tag{$\star$}$$

In particular $f(x_t)$ is non-increasing in t. Convexity at x_t versus x^* gives $f(x_t) - f^* \le \langle \nabla f(x_t), x_t - x^*\rangle$. Expand the gradient step:

$$\|x_{t+1} - x^*\|^2 = \|x_t - x^*\|^2 - \tfrac{2}{L}\langle \nabla f(x_t), x_t - x^*\rangle + \tfrac{1}{L^2}\|\nabla f(x_t)\|^2,$$

so

$$\tfrac{2}{L}\langle \nabla f(x_t), x_t - x^*\rangle = \|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2 + \tfrac{1}{L^2}\|\nabla f(x_t)\|^2,$$

hence

$$f(x_t) - f^* \le \tfrac{L}{2}\big(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2\big) + \tfrac{1}{2L}\|\nabla f(x_t)\|^2.$$

Adding ($\star$) rewritten as $\frac{1}{2L}\|\nabla f(x_t)\|^2 \le f(x_t) - f(x_{t+1})$ to the right-hand side and rearranging,

$$f(x_{t+1}) - f^* \le \tfrac{L}{2}\big(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2\big).$$

Telescope $t = 0, \dots, T-1$:

$$\sum_{t=0}^{T-1}\big(f(x_{t+1}) - f^*\big) \le \tfrac{L}{2}\|x_0 - x^*\|^2.$$

Since $f(x_t)$ is non-increasing, $T(f(x_T) - f^*) \le \sum_{t=0}^{T-1}(f(x_{t+1}) - f^*)$, which yields the claim. □

Theorem 0.62 (GD on L-smooth μ-strongly convex)**.** *Let f be μ-strongly convex and L-smooth, with $\eta = 1/L$. Then*

$$\|x_t - x^*\|^2 \le \big(1 - \mu/L\big)^t\|x_0 - x^*\|^2.$$

Proof. Strong convexity at x_t versus x^* rearranges to

$$\langle \nabla f(x_t), x_t - x^*\rangle \ge f(x_t) - f^* + \tfrac{\mu}{2}\|x_t - x^*\|^2,$$

and from ($\star$) in the previous proof, $\frac{1}{2L}\|\nabla f(x_t)\|^2 \le f(x_t) - f(x_{t+1}) \le f(x_t) - f^*$, i.e. $\frac{1}{L^2}\|\nabla f(x_t)\|^2 \le \frac{2}{L}(f(x_t) - f^*)$. Substituting both into the gradient-step expansion,

$$\begin{aligned}
\|x_{t+1} - x^*\|^2 &= \|x_t - x^*\|^2 - \tfrac{2}{L}\langle \nabla f(x_t), x_t - x^*\rangle + \tfrac{1}{L^2}\|\nabla f(x_t)\|^2 \\
&\le \|x_t - x^*\|^2 - \tfrac{2}{L}\big(f(x_t) - f^* + \tfrac{\mu}{2}\|x_t - x^*\|^2\big) + \tfrac{2}{L}\big(f(x_t) - f^*\big) \\
&= \big(1 - \tfrac{\mu}{L}\big)\|x_t - x^*\|^2.
\end{aligned}$$

Iterate. □

Remark 0.63. The contraction factor $1 - 1/\kappa$ degrades as the condition number $\kappa = L/\mu$ grows: ill-conditioned problems converge slowly. The optimal step $\eta = 2/(L+\mu)$ improves the per-step factor to $((\kappa-1)/(\kappa+1))^2$, but does not change the $O(\kappa\log(1/\varepsilon))$ iteration complexity scaling.

Code sketch

Instantiate $f(x, y) = (x-1)^2 + 2(y+1)^2$, a strongly convex quadratic with Hessian $\mathrm{diag}(2, 4)$, so $\mu = 2$, $L = 4$. Gradient descent with $\eta = 1/L$ contracts $\|x_t - x^*\|^2$ by factor $1 - \mu/L = 1/2$ per step. For a degenerate $f(x) = \|Ax - b\|^2$ with $A \in \mathbb{R}^{20\times 10}$ ill-conditioned, the strong-convexity rate degrades to the convex $O(1/T)$, visible as slope ≈ -1 on a log-log plot of $f(x_T) - f^*$ versus T. A side-by-side run with $\eta = 1/L$ and the optimal $\eta = 2/(L+\mu)$ confirms the sharper contraction predicted by the remark.

Connection to LLMs

Transformer losses are spectacularly non-convex: permuting attention heads or flipping signs in a layernorm gives an equivalent minimum. None of the theorems above apply globally. They apply *locally*, and they supply the operating intuitions that survive the jump.

- **Smooth loss landscape.** The descent lemma says $\eta < 2/L$ is sufficient for monotone decrease. Empirically estimating local L via the Hessian's top eigenvalue is exactly what learning-rate range tests and gradient-norm clipping approximate.
- **Warmup and LR scheduling** (Chapter 27). Early in training the local L is large (sharp minima nearby); a small η avoids the quadratic blow-up term in the descent lemma. As the trajectory enters flatter regions, L drops and η can grow.
- **SGD and AdamW** (Chapters 13, 14). Stochastic gradients break the deterministic descent lemma; the convergence proofs replace $(\star)$ with an expectation inequality plus a variance term, but the algebraic skeleton — descent lemma, telescoping, contraction — is preserved.

The convex theory is a load-bearing analogy: the only place where the rates are honest, and the conceptual scaffold for every heuristic layered on top.

Block B — Probability and Information

Chapter 8: Probability foundations: sample spaces, sigma-algebras, Kolmogorov axioms

Motivation

Every language model is a probability distribution over token sequences. To make rigorous sense of statements like "the model assigns probability p to token t given context c" we need a precise theory of *what* a probability is, *what* it acts on, and *what* consistency rules it must obey. Naive frequentism does not, on its own, justify additivity, well-definedness of conditioning, or continuity in limits. Kolmogorov's 1933 axiomatization solves this by reducing probability to measure theory on a σ-algebra. This chapter builds that machinery from the set-theoretic foundations of Chapter 1 up through inclusion–exclusion, the union bound, Bayes, the law of total probability, and continuity of measure.

Sample spaces, σ-algebras, probability measures

Definition 0.64 (Sample space, event)**.** *A* sample space *is a non-empty set* Ω*. Its elements* $\omega \in \Omega$ *are* outcomes*. An* event *is a subset* $A \subseteq \Omega$*.*

Definition 0.65 (σ-algebra)**.** *A collection $\mathcal{F} \subseteq 2^\Omega$ is a* σ-algebra *on Ω if*

1. $\Omega \in \mathcal{F}$;
2. $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$;
3. $A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^\infty A_n \in \mathcal{F}$.

Remark 0.66. By De Morgan's laws (Chapter 1), $\mathcal{F}$ is closed under countable intersections as well: $\bigcap_n A_n = (\bigcup_n A_n^c)^c$. The closure axioms mirror set algebra exactly, extended to the countably infinite operations probability needs.

Definition 0.67 (Probability measure, probability space)**.** *A function $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a* probability measure *if it satisfies the* Kolmogorov axioms*:*

(K1) $\mathbb{P}(A) \geq 0$ *for all $A \in \mathcal{F}$;*

(K2) $\mathbb{P}(\Omega) = 1$*;*

(K3) *for pairwise disjoint $A_1, A_2, \ldots \in \mathcal{F}$,*

$$\mathbb{P}\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space*.*

Definition 0.68 (Conditional probability, independence)**.** *For $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the* conditional probability *of A given B is*

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Events A, B are independent *iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$.*

Elementary consequences

Lemma 0.69 (Finite additivity, complement, monotonicity)**.** $\mathbb{P}(\emptyset) = 0$*. For disjoint $A, B \in \mathcal{F}$, $\mathbb{P}(A \sqcup B) = \mathbb{P}(A) + \mathbb{P}(B)$. For any A, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.*

Proof. Apply (K3) to $\Omega, \emptyset, \emptyset, \ldots$ (disjoint, with union Ω) to get $1 = 1 + \sum_{n \geq 2} \mathbb{P}(\emptyset)$, so $\mathbb{P}(\emptyset) = 0$. For finite additivity, apply (K3) to $A, B, \emptyset, \emptyset, \ldots$. From $\Omega = A \sqcup A^c$ and (K2), $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$. If $A \subseteq B$, then $B = A \sqcup (B \setminus A)$, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$ by (K1). □

Theorem 0.70 (Inclusion–exclusion, two events)**.** *For all $A, B \in \mathcal{F}$,*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. Decompose into three disjoint pieces:

$$A \cup B = (A \setminus B) \sqcup (B \setminus A) \sqcup (A \cap B), \quad A = (A \setminus B) \sqcup (A \cap B), \quad B = (B \setminus A) \sqcup (A \cap B).$$

By Lemma 0.69,

$$\mathbb{P}(A) + \mathbb{P}(B) = \mathbb{P}(A \setminus B) + \mathbb{P}(B \setminus A) + 2\,\mathbb{P}(A \cap B) = \mathbb{P}(A \cup B) + \mathbb{P}(A \cap B).$$ □

Theorem 0.71 (Union bound / Boole's inequality)**.** *For any countable family* $\{A_n\} \subseteq \mathcal{F}$*,*

$$\mathbb{P}\left(\bigcup_n A_n\right) \le \sum_n \mathbb{P}(A_n).$$

Proof. Disjointify: set $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k<n} A_k$ for $n \ge 2$. Each $B_n \in \mathcal{F}$ (closure under complements and countable unions), the B_n are pairwise disjoint, $B_n \subseteq A_n$, and $\bigsqcup_n B_n = \bigcup_n A_n$. By (K3) and monotonicity,

$$\mathbb{P}\left(\bigcup_n A_n\right) = \sum_n \mathbb{P}(B_n) \le \sum_n \mathbb{P}(A_n).$$ □

Theorem 0.72 (Bayes)**.** *If* $\mathbb{P}(A), \mathbb{P}(B) > 0$*, then*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Proof. By definition, $\mathbb{P}(A \mid B)\,\mathbb{P}(B) = \mathbb{P}(A \cap B) = \mathbb{P}(B \cap A) = \mathbb{P}(B \mid A)\,\mathbb{P}(A)$. Divide by $\mathbb{P}(B)$. □

Theorem 0.73 (Law of total probability)**.** *If* $\{B_i\}_{i \in I}$ *is a countable partition of* Ω *with* $\mathbb{P}(B_i) > 0$*, then for every* $A \in \mathcal{F}$*,*

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i).$$

Proof. $A = A \cap \Omega = A \cap \bigsqcup_i B_i = \bigsqcup_i (A \cap B_i)$, a disjoint union. By (K3), $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$. □

Theorem 0.74 (Continuity of measure)**.** *Let* $A_n \in \mathcal{F}$*.*

1. (From below) *If* $A_1 \subseteq A_2 \subseteq \cdots$ *and* $A = \bigcup_n A_n$*, then* $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$*.*
2. (From above) *If* $A_1 \supseteq A_2 \supseteq \cdots$ *and* $A = \bigcap_n A_n$*, then* $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$*.*

Proof. (Below.) Set $B_1 = A_1$, $B_n = A_n \setminus A_{n-1}$ for $n \ge 2$. The B_n are disjoint, $\bigsqcup_{k \le n} B_k = A_n$, and $\bigsqcup_k B_k = A$. By (K3),

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

(Above.) The sequence $C_n := A_1 \setminus A_n$ is increasing with union $A_1 \setminus A$. By the previous case, $\mathbb{P}(A_1) - \mathbb{P}(A_n) = \mathbb{P}(C_n) \to \mathbb{P}(A_1 \setminus A) = \mathbb{P}(A_1) - \mathbb{P}(A)$. Since $\mathbb{P}(A_1) \le 1 < \infty$, we may subtract: $\mathbb{P}(A_n) \to \mathbb{P}(A)$. □

Code sketch

The companion notebook instantiates $\Omega = \{1, \ldots, 6\}^2$ with $\mathcal{F} = 2^{\Omega}$ and uniform $\mathbb{P}(A) = |A|/36$. It verifies (K2), finite additivity for two disjoint events, two-event inclusion–exclusion, the union bound on three events, an independence vs. non-independence comparison, and a Bayes computation for disease testing checked both analytically and by Monte Carlo.

Connection to LLMs

A causal language model (Chapter 25, forward reference) defines, for each context c, a probability measure on the finite sample space $\Omega = \mathcal{V}$ (the vocabulary), with $\mathcal{F} = 2^{\mathcal{V}}$ and $\mathbb{P}(\{t\} \mid c) = p_\theta(t \mid c)$. The softmax output is precisely a Kolmogorov probability measure: non-negative, normalized, additive on disjoint subsets of tokens. The factorization

$$\mathbb{P}(t_1, \ldots, t_n) = \prod_{i=1}^{n} \mathbb{P}(t_i \mid t_{<i})$$

is iterated conditioning. Sampling, top-k truncation, nucleus filtering, beam search, and importance-weighted training reduce to operations on this measure. Bayes' theorem reappears whenever we invert a generative model into a posterior over latents (alignment, reward modeling). Continuity of measure licenses limiting arguments needed for safety claims about infinite or arbitrarily long generations.

Chapter 9: Random variables, distributions, CDF/PMF/PDF

Motivation

Chapter 8 built probability spaces $(\Omega, \mathcal{F}, \mathbb{P})$ as the bedrock for modeling uncertainty. But raw Ω is rarely what we *measure*: we measure a temperature, a token id, a pixel intensity. A *random variable* is the bridge that turns abstract outcomes into real numbers we can sum, integrate, optimize, and—most importantly—differentiate. Every loss function in deep learning is an expectation over random variables; every sampler in a language model is a draw from a distribution; every softmax is a categorical PMF. This chapter formalizes the machinery.

Definitions

Definition 0.75 (Random variable). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A function $X : \Omega \to \mathbb{R}$ is a* random variable *(RV) if for every $x \in \mathbb{R}$,*

$$\{X \le x\} := \{\omega \in \Omega : X(\omega) \le x\} \in \mathcal{F}.$$

Equivalently, $X^{-1}(B) \in \mathcal{F}$ for every Borel $B \in \mathcal{B}(\mathbb{R})$.

Definition 0.76 (Distribution). *The* distribution *(or* law*) of X is the pushforward measure μ_X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$,*

$$\mu_X(B) = \mathbb{P}(X \in B), \qquad B \in \mathcal{B}(\mathbb{R}).$$

Definition 0.77 (CDF). *The* cumulative distribution function *of X is $F_X : \mathbb{R} \to [0, 1]$,*

$$F_X(x) = \mathbb{P}(X \le x) = \mu_X((-\infty, x]).$$

Definition 0.78 (Discrete RV, PMF). *X is* discrete *if it takes values in a countable set $S \subset \mathbb{R}$. The* probability mass function *is*

$$p_X(x) = \mathbb{P}(X = x), \quad x \in S.$$

Definition 0.79 (Continuous RV, PDF). *X is* (absolutely) continuous *if there exists a non-negative measurable $f_X : \mathbb{R} \to [0, \infty)$, the* probability density function*, with*

$$\mathbb{P}(X \in A) = \int_A f_X(x)\, dx \quad \textit{for every Borel } A \subseteq \mathbb{R}.$$

Example 0.80 (Standard families).

- Bernoulli(p): $p_X(1) = p,\ p_X(0) = 1 - p$.
- Binomial(n, p): $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$.
- Categorical($\pi_1, \ldots, \pi_K$): $p_X(k) = \pi_k$, $\sum_k \pi_k = 1$.
- Geometric(p): $p_X(k) = (1-p)^{k-1} p$, $k \geq 1$.
- Uniform$[a, b]$: $f_X(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x)$.
- Gaussian $\mathcal{N}(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$.

Definition 0.81 (Joint, marginal, conditional)**.** *For (X, Y) on the same space the* joint *distribution is on $\mathbb{R}^2$.* Marginals *integrate/sum out: $f_X(x) = \int f_{X,Y}(x, y)\, dy$. The* conditional *density is $f_{Y|X}(y \mid x) = f_{X,Y}(x, y)/f_X(x)$ when $f_X(x) > 0$.*

Theorems and Proofs

Theorem 0.82 (Properties of CDF)**.** *F_X is (i) non-decreasing, (ii) right-continuous, (iii)* $\lim_{x\to-\infty} F_X(x) = 0$*, (iv)* $\lim_{x\to\infty} F_X(x) = 1$.

Proof. (i) If $x \leq y$ then $\{X \leq x\} \subseteq \{X \leq y\}$, so $\mathbb{P}$-monotonicity gives $F_X(x) \leq F_X(y)$.

(ii) Fix x and let $x_n \downarrow x$. Then $\{X \leq x_n\} \downarrow \{X \leq x\}$ (countable intersection). By continuity of $\mathbb{P}$ from above (Chapter 8, since $\mathbb{P}(\{X \leq x_1\}) \leq 1 < \infty$),

$$F_X(x_n) = \mathbb{P}(X \leq x_n) \to \mathbb{P}(X \leq x) = F_X(x).$$

(iii) Take $x_n \downarrow -\infty$. Then $\{X \leq x_n\} \downarrow \emptyset$, hence $F_X(x_n) \to \mathbb{P}(\emptyset) = 0$.

(iv) Take $x_n \uparrow \infty$. Then $\{X \leq x_n\} \uparrow \Omega$; continuity from below gives $F_X(x_n) \to \mathbb{P}(\Omega) = 1$. □

Theorem 0.83 (Normalization)**.** $\sum_{x\in S} p_X(x) = 1$ *in the discrete case, and* $\int_{\mathbb{R}} f_X(x)\, dx = 1$ *in the continuous case.*

Proof. Discrete: $\{X = x\}$ for $x \in S$ are disjoint with union $\{X \in S\} = \Omega$. Countable additivity:

$$1 = \mathbb{P}(\Omega) = \sum_{x\in S} \mathbb{P}(X = x) = \sum_x p_X(x).$$

Continuous: take $A = \mathbb{R}$ in the defining property: $1 = \mathbb{P}(X \in \mathbb{R}) = \int_{\mathbb{R}} f_X$. □

Theorem 0.84 (Change of variables, 1-D)**.** *Let X have density f_X and let g be strictly monotone and C^1 on the support of X. Then $Y = g(X)$ has density*

$$f_Y(y) = f_X(g^{-1}(y))\, \big|(g^{-1})'(y)\big|.$$

Proof. Suppose g is strictly increasing. For $y \in g(\operatorname{supp} X)$,

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiate using the chain rule (Chapter 3) and the fact that $F_X' = f_X$:

$$f_Y(y) = f_X(g^{-1}(y))\, (g^{-1})'(y).$$

Since g^{-1} is increasing, $(g^{-1})'(y) > 0$, matching the absolute value. If g is decreasing, $\{g(X) \leq y\} = \{X \geq g^{-1}(y)\}$, so $F_Y(y) = 1 - F_X(g^{-1}(y))$ and differentiation gives $-f_X(g^{-1}(y))(g^{-1})'(y)$ with $(g^{-1})'(y) < 0$, again the absolute value. □

Theorem 0.85 (Inverse-CDF sampling)**.** *Let F be a CDF on $\mathbb{R}$ and define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. If $U \sim \mathrm{Uniform}[0,1]$, then $X := F^{-1}(U)$ has CDF F.*

Proof. We claim $\{F^{-1}(U) \leq x\} = \{U \leq F(x)\}$ up to a null set. If $F^{-1}(u) \leq x$, the infimum definition together with right-continuity of F implies $F(x) \geq u$. Conversely, if $u \leq F(x)$ then $x \in \{x' : F(x') \geq u\}$, so $F^{-1}(u) \leq x$. Hence

$$\mathbb{P}(X \leq x) = \mathbb{P}(U \leq F(x)) = F(x),$$

using uniformity of U on $[0,1]$. □

Code Sketch

The notebook (`cells.json`): (1) builds a 5-class categorical via softmax of fixed logits and verifies $\sum p_X = 1$; (2) computes its CDF and samples 10000 draws by inverse-CDF, comparing empirical to true; (3) implements $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, verifies $\int f \approx 1$ by Riemann sum, and computes $F(x)$ by cumulative sum; (4) verifies Theorem 9.3 on $Y = -\ln X$ with $X \sim U[0,1]$, yielding $Y \sim \mathrm{Exp}(1)$; (5) re-runs inverse-CDF sampling on the categorical with growing n to demonstrate convergence.

Connection to LLMs

A causal language model produces logits $z_t \in \mathbb{R}^V$, which softmax maps to a categorical distribution

$$p_\theta(x_t \mid x_{<t}) = \mathrm{softmax}(z_t).$$

Token sampling at temperature τ rescales logits to z_t/τ and draws from this categorical. The standard implementation is exactly inverse-CDF sampling (Theorem 9.4) on the cumulative softmax, or equivalently the *Gumbel-max trick* $\arg\max_k (z_k + G_k)$ with $G_k \sim \mathrm{Gumbel}(0,1)$. Cross-entropy loss (Chapter 17) is $-\log p_\theta(x_t \mid x_{<t})$, an expectation under the data distribution; the causal LM training objective (Chapter 25) is the joint log-likelihood factored through the conditional distributions defined here.

Chapter 10: Expectation, variance, covariance; Jensen's inequality

Motivation

Random variables (Chapter 9) describe uncertain outcomes. To compress an entire distribution into a single representative number we use the *expectation*; to describe spread and joint linear association we use *variance* and *covariance*. Together with *Jensen's inequality*, these objects form the structural backbone of every loss function and convergence proof in modern deep learning. Training an LLM is, formally, the minimization of an expectation $\mathbb{E}_{x \sim \mathcal{D}}[\ell(\theta; x)]$; SGD is a Monte Carlo estimator of that expectation; the variance of the estimator dictates convergence speed (Chapter 13).

Definitions

Definition 0.86 (Expectation)**.** *For a discrete random variable X with pmf p_X,*

$$\mathbb{E}[X] \;=\; \sum_x x\, p_X(x), \qquad \textit{provided} \; \sum_x |x|\, p_X(x) < \infty.$$

For continuous X with density f_X,

$$\mathbb{E}[X] \;=\; \int_{\mathbb{R}} x\, f_X(x)\, dx,$$

provided the integral is absolutely convergent. In full generality, $\mathbb{E}[X] := \int_\Omega X\, d\mathbb{P}$ is the Lebesgue integral against the underlying probability measure (cited without proof; see Billingsley, Probability and Measure*).*

Definition 0.87 (Variance and covariance)**.** *For X with $\mathbb{E}[X^2] < \infty$, $\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}X)^2]$, and $\sigma_X := \sqrt{\mathrm{Var}(X)}$. For X, Y with finite variance,*

$$\mathrm{Cov}(X,Y) := \mathbb{E}\big[(X - \mathbb{E}X)(Y - \mathbb{E}Y)\big], \qquad \rho_{X,Y} := \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

The bound $|\rho| \le 1$ is the Cauchy–Schwarz inequality applied to the inner product $\langle U, V\rangle := \mathbb{E}[UV]$ on the Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 0.88 (Conditional expectation)**.** *For discrete X, Y and any y with $p_Y(y) > 0$,*

$$\mathbb{E}[X \mid Y = y] \;=\; \sum_x x\, p_{X|Y}(x \mid y).$$

Viewed as y varies, $\mathbb{E}[X \mid Y]$ is itself a random variable: a measurable function of Y. The general definition characterizes $\mathbb{E}[X \mid \mathcal{G}]$ as the unique (a.s.) $\mathcal{G}$-measurable random variable satisfying $\int_A \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_A X\, d\mathbb{P}$ for every $A \in \mathcal{G}$ (Radon–Nikodym).

Theorems with proofs

Theorem 0.89 (Linearity of expectation)**.** *For random variables X, Y with finite expectation and scalars $a, b \in \mathbb{R}$,*

$$\mathbb{E}[aX + bY] \;=\; a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

Independence is not *required.*

Proof. In the discrete case, with joint pmf $p_{X,Y}$,

$$\mathbb{E}[aX + bY] = \sum_{x,y} (ax + by)\, p_{X,Y}(x,y) = a \sum_{x,y} x\, p_{X,Y}(x,y) + b \sum_{x,y} y\, p_{X,Y}(x,y).$$

Marginalizing, $\sum_y p_{X,Y}(x,y) = p_X(x)$ and $\sum_x p_{X,Y}(x,y) = p_Y(y)$, giving $a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$. The continuous case is identical with sums replaced by integrals; in full generality, linearity is a defining property of the Lebesgue integral. □

Theorem 0.90 (Variance of a sum)**.**

$$\mathrm{Var}(X + Y) \;=\; \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y).$$

Proof. Let $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$. By linearity, $\mathbb{E}[X + Y] = \mu_X + \mu_Y$. Then

$$\begin{aligned}
\mathrm{Var}(X+Y) &= \mathbb{E}\big[((X - \mu_X) + (Y - \mu_Y))^2\big] \\
&= \mathbb{E}[(X - \mu_X)^2] + \mathbb{E}[(Y - \mu_Y)^2] + 2\,\mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\
&= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y).
\end{aligned}$$

□

Remark 0.91. If $X \perp Y$ then $\mathrm{Cov}(X, Y) = 0$ and variances add. The converse is false: zero covariance does not imply independence.

Theorem 0.92 (Jensen's inequality)**.** *Let $\phi : \mathbb{R} \to \mathbb{R}$ be convex and X a random variable with $\mathbb{E}|X| < \infty$ and $\mathbb{E}|\phi(X)| < \infty$. Then*

$$\phi(\mathbb{E}[X]) \;\le\; \mathbb{E}[\phi(X)].$$

Proof. Recall (Chapter 7) that a convex function on $\mathbb{R}$ admits a *supporting line* at every interior point of its domain: at $x_0 := \mathbb{E}[X]$ there exists a subgradient $g \in \partial\phi(x_0)$ such that

$$\phi(x) \;\ge\; \phi(x_0) + g(x - x_0) \qquad \text{for all } x \in \mathbb{R}.$$

Substitute X pointwise and take expectations. By linearity (Theorem above),

$$\mathbb{E}[\phi(X)] \;\ge\; \phi(x_0) + g\,(\mathbb{E}[X] - x_0) \;=\; \phi(\mathbb{E}[X]),$$

since $\mathbb{E}[X] - x_0 = 0$. □

Theorem 0.93 (Markov's inequality)**.** *Let $X \ge 0$ a.s. and $a > 0$. Then*

$$\mathbb{P}(X \ge a) \;\le\; \frac{\mathbb{E}[X]}{a}.$$

Proof. Pointwise, $a \cdot \mathbf{1}_{\{X \ge a\}} \le X$: when $X \ge a$, both sides are bounded above by X; when $X < a$, the left side is 0 and $X \ge 0$. Take expectations: $a\,\mathbb{P}(X \ge a) \le \mathbb{E}[X]$, then divide by a. □

Theorem 0.94 (Chebyshev's inequality)**.** *Let $\mathbb{E}[X] = \mu$ and $\sigma^2 := \mathrm{Var}(X) < \infty$. For $k > 0$,*

$$\mathbb{P}(|X - \mu| \ge k\sigma) \;\le\; \frac{1}{k^2}.$$

Proof. Apply Markov to $Y := (X - \mu)^2 \ge 0$ with $a = k^2\sigma^2$:

$$\mathbb{P}((X - \mu)^2 \ge k^2\sigma^2) \;\le\; \frac{\mathbb{E}[(X - \mu)^2]}{k^2\sigma^2} \;=\; \frac{1}{k^2}.$$

The event $\{(X - \mu)^2 \ge k^2\sigma^2\}$ equals $\{|X - \mu| \ge k\sigma\}$. □

Theorem 0.95 (Weak law of large numbers)**.** *Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Set $\bar{X}_n := \frac{1}{n}\sum_{i=1}^n X_i$. Then for every $\epsilon > 0$,*

$$\mathbb{P}(|\bar{X}_n - \mu| \ge \epsilon) \xrightarrow{n \to \infty} 0.$$

Proof. By linearity, $\mathbb{E}[\bar{X}_n] = \mu$. By the variance-of-a-sum theorem and independence (covariances vanish), $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev's inequality yields

$$\mathbb{P}(|\bar{X}_n - \mu| \ge \epsilon) \;\le\; \frac{\mathrm{Var}(\bar{X}_n)}{\epsilon^2} \;=\; \frac{\sigma^2}{n\epsilon^2} \;\to\; 0.$$ □

Code sketch

The companion notebook (`cells.json`) verifies each result numerically: a five-outcome categorical for analytic vs. empirical $\mathbb{E}, \mathrm{Var}$; the pair $(X, 2X + 1)$ for linearity without independence; $\phi(x) = e^x$ on $\mathrm{Unif}\{-1, +1\}$ for Jensen; and i.i.d. $\mathrm{Unif}[0, 1]$ samples for the LLN inside a Chebyshev confidence band.

Connection to LLMs

Every modern language-model objective has the form

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{x\sim\mathcal{D}}[\ell(\theta;x)],$$

where $\mathcal{D}$ is the data distribution and ℓ is (typically) the next-token cross-entropy loss. The data distribution is intractable, so we replace the expectation by an empirical mean over a mini-batch of size B:

$$\hat{\mathcal{L}}(\theta) \;=\; \frac{1}{B}\sum_{i=1}^{B}\ell(\theta;x_i).$$

Linearity of expectation gives $\mathbb{E}[\nabla\hat{\mathcal{L}}] = \nabla\mathcal{L}$, so the SGD gradient is *unbiased*. Theorem on variance of a sum, plus independence within the batch, yields $\mathrm{Var}(\nabla\hat{\mathcal{L}}) = \Theta(1/B)$ — exactly the reason larger batches give smoother training curves. The weak LLN guarantees recovery of the true loss in probability as $B\to\infty$ (formalized in Chapter 13). Jensen's inequality, applied to $-\log$, underwrites the evidence lower bound (ELBO) of Chapter 22:

$$\log\mathbb{E}_q[Z] \;\geq\; \mathbb{E}_q[\log Z], \qquad Z>0,$$

which is the bedrock of every modern likelihood-based generative model.

Chapter 11: Information theory: self-information, entropy, cross-entropy, KL

Motivation

Chapter 9 gave us random variables; Chapter 10 gave us expectation and Jensen's inequality. We now ask: how much *information* does an outcome carry, and how *different* are two distributions on the same space? The answers — entropy, cross-entropy, and the Kullback–Leibler divergence — are not optional flavor. They *are* the loss surface of every modern language model. Cross-entropy is the next-token training objective (Chapters 17, 25); KL is the trust-region regularizer in RLHF / DPO (Chapter 28); the entropy of the softmax is what people mean by "sampling diversity." Every result below follows from one fact already proved: $-\ln$ is convex.

We use the natural log ln throughout for clean derivatives. Replacing ln by $\log_2$ multiplies every formula by $1/\ln 2$ and converts *nats* to *bits*; nothing else changes.

Definitions

Definition 0.96 (Self-information). *For a discrete random variable X with PMF p and $x\in\mathrm{supp}(p)$, the* self-information *of the outcome x is $I(x) := -\ln p(x)$. Rare outcomes carry more information; $I(x)\to\infty$ as $p(x)\to 0$.*

Definition 0.97 (Shannon entropy). *The* entropy *of p is the expected self-information,*

$$H(p) := \mathbb{E}_{x\sim p}[I(x)] = -\sum_x p(x)\ln p(x),$$

with the convention $0\ln 0 = 0$, justified by $\lim_{t\downarrow 0} t\ln t = 0$.

Definition 0.98 (Joint and conditional entropy)**.** *For a joint PMF $p_{X,Y}$,*

$$H(X,Y) = -\sum_{x,y} p_{X,Y}(x,y)\ln p_{X,Y}(x,y), \qquad H(Y \mid X) = -\sum_{x,y} p_{X,Y}(x,y)\ln p_{Y|X}(y \mid x).$$

Definition 0.99 (Cross-entropy)**.** *For PMFs p, q on the same alphabet with* $\operatorname{supp}(p) \subseteq \operatorname{supp}(q)$,

$$H(p,q) := -\sum_x p(x)\ln q(x) = \mathbb{E}_{x\sim p}[-\ln q(x)].$$

Definition 0.100 (Kullback–Leibler divergence)**.**

$$D_{\mathrm{KL}}(p\,\|\,q) := \sum_x p(x)\ln\frac{p(x)}{q(x)} = \mathbb{E}_{x\sim p}\left[\ln\frac{p(x)}{q(x)}\right].$$

D_{KL} *is* not *symmetric and* not *a metric.*

Definition 0.101 (Mutual information)**.** *For joint $p_{X,Y}$ with marginals p_X, p_Y,*

$$I(X;Y) := D_{\mathrm{KL}}(p_{X,Y}\,\|\,p_X \otimes p_Y) = \sum_{x,y} p_{X,Y}(x,y)\ln\frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)}.$$

Theorems and proofs

Theorem 0.102 (Gibbs' inequality / KL nonnegativity)**.** *For PMFs p, q on the same finite alphabet with $q(x) > 0$ wherever $p(x) > 0$,*

$$D_{\mathrm{KL}}(p\,\|\,q) \ge 0,$$

with equality iff $p(x) = q(x)$ for every x with $p(x) > 0$.

Proof. The function $\phi(t) = -\ln t$ is strictly convex on $(0,\infty)$ since $\phi''(t) = 1/t^2 > 0$. Compute, viewing the sum as $\mathbb{E}_{x\sim p}$:

$$-D_{\mathrm{KL}}(p\,\|\,q) = \sum_{x:p(x)>0} p(x)\ln\frac{q(x)}{p(x)} = -\,\mathbb{E}_{x\sim p}\left[\phi\Big(\tfrac{q(x)}{p(x)}\Big)\right].$$

By Jensen's inequality (Chapter 10), $\mathbb{E}[\phi(Z)] \ge \phi(\mathbb{E}[Z])$ for convex ϕ, hence

$$-D_{\mathrm{KL}}(p\,\|\,q) \le -\phi\Big(\mathbb{E}_{x\sim p}\Big[\tfrac{q(x)}{p(x)}\Big]\Big) = \ln\Big(\sum_{x:p(x)>0} q(x)\Big) \le \ln 1 = 0,$$

using $\sum_x q(x) = 1$. Therefore $D_{\mathrm{KL}}(p\,\|\,q) \ge 0$. By strict convexity, Jensen is an equality iff $q(x)/p(x)$ is p-a.s. constant; since both p and q sum to 1 on the support of p, that constant must be 1, i.e. $p = q$ on $\operatorname{supp}(p)$. □

Theorem 0.103 (Cross-entropy decomposition)**.** $H(p,q) = H(p) + D_{\mathrm{KL}}(p\,\|\,q)$.

Proof. Direct algebra:

$$H(p,q) = -\sum_x p(x)\ln q(x) = -\sum_x p(x)\big[\ln p(x) + \ln\tfrac{q(x)}{p(x)}\big] = H(p) + D_{\mathrm{KL}}(p\,\|\,q).$$ □

Corollary 0.104. *$H(p,q) \ge H(p)$, with equality iff $q = p$. In particular, $\min_q H(p,q) = H(p)$, attained at $q = p$.*

Theorem 0.105 (Maximum entropy on a finite alphabet)**.** *For any PMF p supported on a set of size K,*

$$H(p) \le \ln K,$$

with equality iff p is the uniform distribution $u(x) = 1/K$.

Proof. Let u denote the uniform distribution on the K-element support. Compute

$$D_{\mathrm{KL}}(p \,\|\, u) = \sum_x p(x) \ln \frac{p(x)}{1/K} = -H(p) + \ln K \cdot \sum_x p(x) = \ln K - H(p).$$

By Gibbs' inequality this is ≥ 0, so $H(p) \le \ln K$, with equality iff $p = u$. □

Theorem 0.106 (Chain rule for entropy)**.** $H(X, Y) = H(X) + H(Y \mid X)$.

Proof. Apply $\ln p_{X,Y}(x, y) = \ln p_X(x) + \ln p_{Y|X}(y \mid x)$:

$$H(X, Y) = -\sum_{x,y} p_{X,Y}(x, y) \ln p_X(x) \; - \; \sum_{x,y} p_{X,Y}(x, y) \ln p_{Y|X}(y \mid x).$$

Marginalizing y in the first sum gives $-\sum_x p_X(x) \ln p_X(x) = H(X)$; the second sum is $H(Y \mid X)$ by definition. □

Theorem 0.107 (Mutual information nonnegativity)**.** $I(X; Y) \ge 0$, *with equality iff X and Y are independent.*

Proof. By definition $I(X; Y) = D_{\mathrm{KL}}(p_{X,Y} \,\|\, p_X \otimes p_Y)$. Apply Gibbs (Theorem 1): this is ≥ 0 with equality iff $p_{X,Y}(x, y) = p_X(x) p_Y(y)$ for all x, y — exactly independence. □

Remark 0.108. Combining the chain rule with the definition of I yields the standard symmetric identity

$$I(X; Y) = H(X) + H(Y) - H(X, Y) = H(Y) - H(Y \mid X) = H(X) - H(X \mid Y).$$

Conditioning never increases entropy: $H(Y \mid X) \le H(Y)$.

Code sketch

The notebook builds two categorical distributions p, q on a 5-token alphabet, implements `entropy`, `cross_entropy`, and `kl` from the definitions above, and verifies $H(p, q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$ to machine precision. It samples 50 random Dirichlet pairs (seed 0) and confirms $D_{\mathrm{KL}} \ge 0$ for every pair, with $D_{\mathrm{KL}}(p \,\|\, p) = 0$. For $K = 8$, it confirms $H(\text{uniform}) = \ln 8$ and that $H(p) < \ln 8$ for 100 non-uniform draws. Finally it constructs a correlated joint on $\{0, 1\}^2$, computes $I(X; Y)$ via both formulas, verifies agreement, and shows $I = 0$ for the product distribution.

Connection to LLMs

A causal language model defines $p_\theta(x_t \mid x_{<t})$ via softmax (Chapter 9). On a corpus drawn from p_{data}, the training objective is

$$\mathcal{L}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}}[-\ln p_\theta(x_t \mid x_{<t})] = H(p_{\text{data}}, p_\theta) = H(p_{\text{data}}) + D_{\mathrm{KL}}(p_{\text{data}} \,\|\, p_\theta).$$

Theorem 2 makes precise why minimizing cross-entropy *is* minimizing KL to the data: the entropy term $H(p_{\text{data}})$ does not depend on θ. Theorem 4 sets the absolute floor: a uniform LM over a

50000-token vocabulary has entropy $\ln 50000 \approx 10.82$ nats, which is the loss every trained model improves on (Chapter 17, 25). In RLHF and DPO (Chapter 28), the policy update is regularized by $D_{\mathrm{KL}}(\pi_\theta \,\|\, \pi_{\mathrm{ref}})$ to keep the fine-tuned policy close to the SFT initialization — Gibbs' inequality is exactly what makes that penalty a meaningful divergence. Finally, the entropy of the sampler's softmax, controlled by temperature, is the "diversity" knob in decoding.

Chapter 12: Statistical inference: likelihood, MLE, ERM, bias-variance

Motivation

Probability theory (Chapters 7–10) reasons forward from a known distribution to data. *Statistical inference* runs the arrow backwards: given data, recover the distribution. Two principles dominate modern practice and underwrite essentially all of deep learning: *maximum likelihood estimation* (MLE) and its generalization, *empirical risk minimization* (ERM). We will show that MLE is precisely the minimizer of cross-entropy between the empirical distribution and the model, connecting Chapter 11's information theory to Chapter 17's language-model objective.

Definitions

Definition 0.109 (Statistical model and iid sample)**.** *A* statistical model *is a family $\mathcal{P} = \{p_\theta : \theta \in \Theta\}$ of probability densities (or mass functions) on a sample space $\mathcal{X}$, indexed by $\theta \in \Theta \subseteq \mathbb{R}^d$. A sample $X_1, \dots, X_n$ is* iid *from p_{θ^*} if the joint density factors as $\prod_{i=1}^n p_{\theta^*}(x_i)$.*

Definition 0.110 (Likelihood and MLE)**.** *The* likelihood *and* log-likelihood *are*

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n p_\theta(x_i), \qquad \ell(\theta; \mathbf{x}) = \sum_{i=1}^n \log p_\theta(x_i).$$

The maximum likelihood estimator *is $\hat\theta_{\mathrm{MLE}} = \arg\max_{\theta\in\Theta} \ell(\theta; \mathbf{x})$.*

Definition 0.111 (Empirical risk minimization)**.** *Given a loss $\ell(\theta; x)$ (not necessarily $-\log p_\theta$),*

$$\hat\theta_{\mathrm{ERM}} = \arg\min_{\theta\in\Theta} \frac{1}{n} \sum_{i=1}^n \ell(\theta; x_i).$$

MLE is ERM with $\ell(\theta; x) = -\log p_\theta(x)$.

Definition 0.112 (Bias, variance, MSE)**.** *For an estimator $\hat\theta = \hat\theta(X_1, \dots, X_n)$ of a scalar θ^*:*

$$\mathrm{Bias}(\hat\theta) = \mathbb{E}[\hat\theta] - \theta^*, \quad \mathrm{Var}(\hat\theta) = \mathbb{E}[(\hat\theta - \mathbb{E}\hat\theta)^2], \quad \mathrm{MSE}(\hat\theta) = \mathbb{E}[(\hat\theta - \theta^*)^2].$$

Theorems

Theorem 0.113 (MLE equals minimum cross-entropy)**.** *Let $\hat p_n(x) = \frac{1}{n}\sum_{i=1}^n \mathbf{1}\{x_i = x\}$ be the empirical distribution. Then*

$$\arg\max_\theta \ell(\theta; \mathbf{x}) \;=\; \arg\min_\theta H(\hat p_n,\, p_\theta),$$

where $H(\hat p_n, p_\theta) = -\sum_x \hat p_n(x) \log p_\theta(x)$ is the cross-entropy from Chapter 11.

Proof. Direct algebra:

$$\tfrac{1}{n}\,\ell(\theta) = \tfrac{1}{n}\sum_{i=1}^{n}\log p_\theta(x_i) = \sum_x \hat{p}_n(x)\log p_\theta(x) = -H(\hat{p}_n, p_\theta).$$

Maximizing ℓ is equivalent to maximizing ℓ/n, which is equivalent to minimizing $H(\hat{p}_n, p_\theta)$. ☐

Corollary 0.114 (MLE equals minimum KL)**.** *By the Chapter 11 identity $H(\hat{p}_n, p_\theta) = H(\hat{p}_n) + D_{\mathrm{KL}}(\hat{p}_n \,\|\, p_\theta)$, and since $H(\hat{p}_n)$ does not depend on θ,*

$$\hat{\theta}_{\mathrm{MLE}} = \arg\min_\theta D_{\mathrm{KL}}(\hat{p}_n \,\|\, p_\theta).$$

Theorem 0.115 (Bias–variance decomposition)**.** *For any square-integrable estimator $\hat{\theta}$ of $\theta^* \in \mathbb{R}$,*

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

Proof. Let $\bar{\theta} = \mathbb{E}[\hat{\theta}]$. Add and subtract $\bar{\theta}$ inside the square:

$$\mathbb{E}[(\hat{\theta} - \theta^*)^2] = \mathbb{E}\big[((\hat{\theta} - \bar{\theta}) + (\bar{\theta} - \theta^*))^2\big].$$

Expand:

$$= \mathbb{E}[(\hat{\theta} - \bar{\theta})^2] + 2(\bar{\theta} - \theta^*)\,\mathbb{E}[\hat{\theta} - \bar{\theta}] + (\bar{\theta} - \theta^*)^2.$$

The cross term vanishes because $\mathbb{E}[\hat{\theta} - \bar{\theta}] = 0$, leaving $\mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2$. ☐

Theorem 0.116 (MLE for Gaussian mean, known variance)**.** *Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known. Then $\hat{\mu}_{\mathrm{MLE}} = \bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$.*

Proof. The log-likelihood is

$$\ell(\mu) = -\tfrac{n}{2}\log(2\pi\sigma^2) - \tfrac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Differentiate:

$$\ell'(\mu) = \tfrac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu).$$

Setting $\ell'(\mu) = 0$ gives $\sum_i X_i = n\mu$, i.e. $\hat{\mu} = \bar{X}_n$. The second derivative is $-n/\sigma^2 < 0$, so this is a strict maximum. ☐

Theorem 0.117 (Consistency of MLE, sketch)**.** *Under regularity conditions (identifiability, compact Θ, dominated $\log p_\theta$), $\hat{\theta}_n \xrightarrow{P} \theta^*$ as $n \to \infty$.*

Sketch. By the Chapter 10 LLN, $\frac{1}{n}\ell_n(\theta) \xrightarrow{P} M(\theta) := \mathbb{E}_{\theta^*}[\log p_\theta(X)]$. Gibbs' inequality (Chapter 11) gives $M(\theta) \le M(\theta^*)$ with equality iff $p_\theta = p_{\theta^*}$ a.s., so θ^* is the unique maximizer of M. A uniform LLN over Θ transfers the maximizer of ℓ_n/n to that of M, yielding $\hat{\theta}_n \xrightarrow{P} \theta^*$. ☐

Code sketch

The accompanying notebook (`cells.json`): (i) simulates $n = 1000$ samples from $\mathcal{N}(2, 1)$, evaluates $\ell(\mu)$ on a grid, and confirms $\arg\max = \bar{X}_n$; (ii) draws $n = 500$ from a categorical with $p = (0.1, 0.2, 0.3, 0.4)$, computes $\hat{p}_n$, and checks that $H(\hat{p}_n, p_\theta)$ is minimized exactly at $\theta = \hat{p}_n$; (iii) empirically decomposes MSE for two estimators of a Gaussian mean; (iv) solves linear regression as ERM via the normal equation.

Connection to LLMs

Remark 0.118. Language-model *pre-training* is MLE. For an autoregressive model, $p_\theta(x_{1:T}) = \prod_t p_\theta(x_t \mid x_{<t})$, so the corpus log-likelihood is $\sum_{\text{seq}} \sum_t \log p_\theta(x_t \mid x_{<t})$. The training loss $-\ell(\theta)/N$ is exactly the cross-entropy $H(\hat{p}_n, p_\theta)$ between the empirical token distribution and the model. By Theorem 0.113 and its corollary, every gradient step of a transformer (Chapter 25) is one step of MLE = ERM with log-loss = KL projection toward the empirical corpus. Bias–variance reasoning (Theorem 0.115) then governs the scaling laws and generalization gap discussed in Chapter 27.

Block C — Stochastic Optimization

Chapter 13: SGD: stochastic-approximation theorem; mini-batching; convergence sketch

Motivation

Chapter 7 proved that gradient descent on an L-smooth convex function reaches an ϵ-stationary point in $O(1/\epsilon)$ iterations using the *full* gradient $\nabla F(\theta)$. For a transformer pre-trained on 10^{13} tokens, a single full-gradient step would require a forward/backward pass over the entire corpus — weeks of compute per update. The fix is to estimate ∇F from a random *mini-batch* and accept a noisy step in exchange for a cheap one. This chapter justifies the trade. We define the stochastic objective, prove that mini-batching reduces variance as $1/B$, and combine the descent lemma of Chapter 7 with the linearity of expectation (Chapter 10) to derive the canonical $O(1/\sqrt{T})$ rate of stochastic gradient descent on smooth (possibly non-convex) losses.

Definitions

Definition 0.119 (Stochastic objective). *Let $\xi \sim \mathcal{D}$ be a random data point and $f(\cdot;\xi) : \mathbb{R}^n \to \mathbb{R}$ a loss. The* population risk *is*

$$F(\theta) := \mathbb{E}_{\xi\sim\mathcal{D}}[f(\theta;\xi)],$$

in the sense of Chapter 10. Training data are i.i.d. draws $\xi_1, \xi_2, \ldots$ from $\mathcal{D}$.

Definition 0.120 (Stochastic gradient). $\hat{g}(\theta;\xi) := \nabla_\theta f(\theta;\xi)$. *Under mild regularity (interchange of ∇ and $\mathbb{E}$),*

$$\mathbb{E}_\xi[\hat{g}(\theta;\xi)] = \nabla F(\theta) \qquad (\textit{unbiasedness}).$$

Definition 0.121 (SGD update). *Given step size $\eta_t > 0$ and an independent sample $\xi_t \sim \mathcal{D}$,*

$$\theta_{t+1} = \theta_t - \eta_t \hat{g}(\theta_t;\xi_t).$$

Definition 0.122 (Mini-batch SGD). *Draw B i.i.d. samples $\xi_1, \ldots, \xi_B$ and average:*

$$\hat{g}_B(\theta) := \frac{1}{B}\sum_{j=1}^{B} \nabla_\theta f(\theta;\xi_j).$$

Definition 0.123 (Bounded variance). $\mathbb{E}_\xi\|\hat{g}(\theta;\xi) - \nabla F(\theta)\|^2 \le \sigma^2$ *uniformly in θ.*

Theorems and proofs

Theorem 0.124 (Variance reduction by batching)**.** *Under bounded variance and i.i.d. sampling,*

$$\mathbb{E}\|\hat{g}_B(\theta) - \nabla F(\theta)\|^2 \le \frac{\sigma^2}{B}.$$

Proof. Let $Z_j := \hat{g}(\theta;\xi_j) - \nabla F(\theta)$. The Z_j are i.i.d., zero-mean, with $\mathbb{E}\|Z_j\|^2 \le \sigma^2$. Then $\hat{g}_B - \nabla F = \frac{1}{B}\sum_j Z_j$, and

$$\mathbb{E}\Big\|\tfrac{1}{B}\sum_j Z_j\Big\|^2 = \tfrac{1}{B^2}\sum_{j,k}\mathbb{E}\langle Z_j, Z_k\rangle = \tfrac{1}{B^2}\sum_j \mathbb{E}\|Z_j\|^2 \le \frac{\sigma^2}{B},$$

where cross terms vanish by independence and zero mean (Theorem 10.6). □

Remark 0.125*.* This is the law of large numbers (Chapter 10) applied to the gradient: doubling the batch halves the noise variance.

Theorem 0.126 (SGD on L-smooth, possibly non-convex F)**.** *Suppose F is L-smooth, bounded below by $F^\star$, and run SGD with constant step $\eta = \min(1/L,\, c/\sqrt{T})$ for T iterations, with stochastic gradients of variance at most σ^2/B. Then*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(\theta_t)\|^2 \le \frac{2\big(F(\theta_0) - F^\star\big)}{\eta T} + \frac{L\eta\sigma^2}{B} = O(1/\sqrt{T}).$$

Proof. Let $g_t := \nabla F(\theta_t)$ and write $\hat{g}_t = g_t + \zeta_t$ with $\mathbb{E}[\zeta_t \mid \theta_t] = 0$ and $\mathbb{E}\|\zeta_t\|^2 \le \sigma^2/B$. The descent lemma of Chapter 7, applied to the displacement $\theta_{t+1} - \theta_t = -\eta\hat{g}_t$, gives

$$F(\theta_{t+1}) \le F(\theta_t) - \eta\langle g_t, \hat{g}_t\rangle + \tfrac{L\eta^2}{2}\|\hat{g}_t\|^2.$$

Take conditional expectation given θ_t. Using $\mathbb{E}[\hat{g}_t \mid \theta_t] = g_t$ and $\mathbb{E}\|\hat{g}_t\|^2 = \|g_t\|^2 + \mathbb{E}\|\zeta_t\|^2$,

$$\mathbb{E}[F(\theta_{t+1}) \mid \theta_t] \le F(\theta_t) - \eta\|g_t\|^2 + \tfrac{L\eta^2}{2}\big(\|g_t\|^2 + \sigma^2/B\big).$$

With $\eta \le 1/L$ we have $\frac{L\eta^2}{2} \le \eta/2$, so the coefficient of $\|g_t\|^2$ becomes $-\eta + \eta/2 = -\eta/2$:

$$\mathbb{E}[F(\theta_{t+1}) \mid \theta_t] \le F(\theta_t) - \tfrac{\eta}{2}\|g_t\|^2 + \tfrac{L\eta^2\sigma^2}{2B}.$$

Take the full expectation, sum $t = 0, \dots, T-1$, and telescope:

$$\tfrac{\eta}{2}\sum_{t=0}^{T-1}\mathbb{E}\|g_t\|^2 \le F(\theta_0) - \mathbb{E}F(\theta_T) + \tfrac{L\eta^2\sigma^2 T}{2B} \le F(\theta_0) - F^\star + \tfrac{L\eta^2\sigma^2 T}{2B}.$$

Divide by $\eta T/2$:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|g_t\|^2 \le \frac{2(F(\theta_0) - F^\star)}{\eta T} + \frac{L\eta\sigma^2}{B}.$$

Choose $\eta = c/\sqrt{T}$ (subject to $\eta \le 1/L$); both right-hand terms become $O(1/\sqrt{T})$, with $c = \sqrt{2(F(\theta_0) - F^\star)B/(L\sigma^2)}$ minimizing the bound. □

Remark 0.127*.* The average squared gradient norm decays like $1/\sqrt{T}$ and the noise floor is proportional to $\eta\sigma^2/B$. Larger batches permit a larger η at the same noise level.

Theorem 0.128 (SGD on L-smooth, μ-strongly convex F; sketch)**.** *With diminishing step $\eta_t = \frac{2}{\mu(t+t_0)}$ for t_0 large enough that $\eta_0 \le 1/L$,*

$$\mathbb{E}\|\theta_T - \theta^\star\|^2 \le \frac{C}{T}.$$

Sketch. Let $r_t^2 := \mathbb{E}\|\theta_t - \theta^\star\|^2$. Expand $\|\theta_{t+1} - \theta^\star\|^2 = \|\theta_t - \theta^\star - \eta_t \hat{g}_t\|^2$ and take expectation. Strong convexity gives $\langle g_t, \theta_t - \theta^\star \rangle \ge \mu\|\theta_t - \theta^\star\|^2$ (Chapter 7), and the noise contributes $\eta_t^2\sigma^2/B$:

$$r_{t+1}^2 \le (1 - \eta_t\mu)\, r_t^2 + \eta_t^2\sigma^2/B.$$

With $\eta_t\mu = 2/(t+t_0)$, induction $r_t^2 \le C/(t+t_0)$ closes for $C \ge 4\sigma^2/(\mu^2 B)$, giving $r_T^2 = O(1/T)$. □

Proposition 0.129 (Robbins–Monro conditions, 1951)**.** *For $\theta_{t+1} = \theta_t - \eta_t\hat{g}_t$ to converge almost surely to a stationary point of F under standard regularity, the step sizes must satisfy*

$$\sum_{t=0}^{\infty} \eta_t = \infty, \qquad \sum_{t=0}^{\infty} \eta_t^2 < \infty.$$

Remark 0.130. $\sum \eta_t = \infty$ guarantees the iterates travel far enough to escape any bounded region; $\sum \eta_t^2 < \infty$ controls the cumulative noise $\sum \eta_t \zeta_t$ in second moment (martingale-convergence). The schedule $\eta_t = c/(t+1)$ satisfies both; $\eta_t = c/\sqrt{t+1}$ fails square-summability; a constant step satisfies neither.

Code sketch

The accompanying notebook compares full-batch GD against SGD on a 1-D least-squares loss; empirically verifies $\mathrm{Var}(\hat{g}_B) \propto 1/B$ for $B \in \{1, 8, 64, 256\}$; runs SGD on the smooth non-convex toy $F(\theta) = \theta^2/2 + 0.5\sin(5\theta)$ to display $T^{-1/2}$ decay of the running average of $\|\nabla F(\theta_t)\|^2$; and contrasts a constant step (which plateaus at a noise floor) with $\eta_t = c/(t+1)$ (which converges).

Connection to LLMs

Pre-training a language model is mini-batch SGD on the cross-entropy loss (Chapter 11) of next-token prediction. The "batch size" of scaling-law papers is exactly the B of Theorem 13.1; doubling it halves the per-step gradient variance but doubles the FLOPs per step. The $O(1/\sqrt{T})$ rate of Theorem 13.2 is why pre-training, even of huge models, must consume *many* tokens — there is no $1/T$ shortcut without strong convexity. In practice we use AdamW (Chapter 14), which adds momentum and per-coordinate adaptive scaling on top of the SGD skeleton derived here; the convergence intuition (variance reduction by batching, noise floor $\propto \eta\sigma^2/B$, $1/\sqrt{T}$ asymptotics) carries over essentially unchanged.

Chapter 14: Momentum, RMSProp, AdamW: derivation and bias-correction proof

Motivation

Chapter 13 established that stochastic gradient descent (SGD) converges, but its rate is governed by the condition number $\kappa = L/\mu$ of the loss landscape. For deep networks — and for transformer language models in particular — κ can be astronomical, with curvature varying by many orders of magnitude across coordinates. Plain SGD wastes most of its updates fighting oscillation in stiff directions while crawling along flat ones. Chapter 14 builds the optimizers that fix both pathologies: *momentum* smooths the trajectory, *RMSProp* rescales per coordinate, *Adam* combines both with bias correction, and *AdamW* decouples weight decay so it survives Adam's rescaling. AdamW is the optimizer behind essentially every transformer pre-training run from GPT-2 onward (forward reference: Chapter 27).

Definitions

Definition 0.131 (Polyak's heavy-ball momentum). *Given gradient estimate $g_t = \nabla f(\theta_t; \xi_t)$ and momentum coefficient $\beta \in [0, 1)$,*

$$v_{t+1} = \beta v_t + g_t, \qquad \theta_{t+1} = \theta_t - \eta\, v_{t+1},$$

with $v_0 = 0$. The state v_t accumulates an exponentially weighted history of past gradients.

Definition 0.132 (Nesterov accelerated gradient). *NAG evaluates the gradient at the look-ahead point $\theta_t - \eta\beta v_t$ rather than at θ_t. Empirically it improves the constant factor on convex problems; in deep learning practice it is a minor variation on heavy-ball.*

Definition 0.133 (RMSProp). *With $\beta_2 \in [0, 1)$ and small $\varepsilon > 0$,*

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)\, g_t^{\odot 2}, \qquad \theta_{t+1} = \theta_t - \eta\, g_t \oslash (\sqrt{v_t} + \varepsilon),$$

where $\odot, \oslash$ are element-wise. RMSProp normalizes each coordinate by an EMA of its squared gradient.

Definition 0.134 (Adam).

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^{\odot 2}, \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}, \\ \theta_{t+1} &= \theta_t - \eta\, \hat{m}_t \oslash (\sqrt{\hat{v}_t} + \varepsilon). \end{aligned}$$

Defaults: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$.

Definition 0.135 (AdamW: decoupled weight decay).

$$\theta_{t+1} = \theta_t - \eta\Big(\hat{m}_t \oslash (\sqrt{\hat{v}_t} + \varepsilon) + \lambda\, \theta_t\Big).$$

The decay term is applied directly to the parameters and is not *funneled through the $\hat{v}_t$ rescaling.*

Theorems and proofs

Lemma 0.136 (EMA closed form). *With $m_0 = 0$, the recursion $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$ unrolls to*

$$m_t = (1 - \beta_1) \sum_{i=1}^{t} \beta_1^{t-i}\, g_i.$$

Proof. Induction on t. Base $t = 1$: $m_1 = (1-\beta_1)g_1$. Step: assume the formula for $t-1$; then

$$m_t = \beta_1\Big[(1-\beta_1)\sum_{i=1}^{t-1}\beta_1^{t-1-i}g_i\Big] + (1-\beta_1)g_t$$
$$= (1-\beta_1)\Big[\sum_{i=1}^{t-1}\beta_1^{t-i}g_i + g_t\Big] = (1-\beta_1)\sum_{i=1}^{t}\beta_1^{t-i}g_i. \qquad \square$$

Theorem 0.137 (Bias correction for Adam, first moment)**.** *Assume g_t is stationary with $\mathbb{E}[g_t] = g$. Then $\mathbb{E}[m_t] = (1-\beta_1^t)\, g$, hence $\mathbb{E}[\hat{m}_t] = g$ for every $t \geq 1$.*

Proof. By the lemma and linearity of expectation,

$$\mathbb{E}[m_t] = (1-\beta_1)\sum_{i=1}^{t}\beta_1^{t-i}\,\mathbb{E}[g_i] = (1-\beta_1)\, g\sum_{j=0}^{t-1}\beta_1^j = (1-\beta_1)\, g\,\frac{1-\beta_1^t}{1-\beta_1} = (1-\beta_1^t)\, g.$$

Dividing by $1-\beta_1^t$ yields $\mathbb{E}[\hat{m}_t] = g$. $\square$

The identical argument with $g_t^{\odot 2}$ in place of g_t shows $\mathbb{E}[\hat{v}_t] = \mathbb{E}[g_t^{\odot 2}]$ under stationarity. Without bias correction, $m_1 = (1-\beta_1)g_1 \approx 0.1\, g$ at the default $\beta_1 = 0.9$ — a tenfold underestimate that would cripple the first hundred or so steps of training.

Proposition 0.138 (Momentum as a low-pass filter)**.** *For heavy-ball with $v_0 = 0$, $v_t = \sum_{i=1}^{t}\beta^{\,t-i}g_i$. The effective averaging horizon is $\sum_{j=0}^{\infty}\beta^j = 1/(1-\beta)$; e.g. $\beta = 0.9$ gives a horizon of about 10 steps. High-frequency gradient noise is attenuated; the consistent low-frequency signal accumulates.*

Proof. Identical unrolling to the Adam lemma but without the $(1-\beta)$ prefactor. The geometric series gives $\sum_{j=0}^{t-1}\beta^j \to 1/(1-\beta)$. $\square$

Theorem 0.139 (Heavy-ball acceleration on convex quadratics, sketch)**.** *For $f(\theta) = \frac{1}{2}\theta^\top A\theta$ with $\mu I \preceq A \preceq LI$ and $\kappa = L/\mu$, GD requires $O(\kappa \log\frac{1}{\epsilon})$ steps to reach ϵ-accuracy, while heavy-ball with optimally tuned η, β achieves $O(\sqrt{\kappa}\log\frac{1}{\epsilon})$.*

Sketch. Diagonalize $A = U\Lambda U^\top$ and decompose θ_t in eigen-coordinates. In an eigen-direction with eigenvalue $\lambda \in [\mu, L]$ the heavy-ball recurrence reads

$$\begin{pmatrix} c_{t+1} \\ c_t \end{pmatrix} = M_\lambda \begin{pmatrix} c_t \\ c_{t-1} \end{pmatrix}, \qquad M_\lambda = \begin{pmatrix} 1+\beta-\eta\lambda & -\beta \\ 1 & 0 \end{pmatrix}.$$

The convergence rate is $\max_{\lambda\in[\mu,L]}\rho(M_\lambda)$, where ρ is the spectral radius. The eigenvalues of M_λ satisfy $z^2 - (1+\beta-\eta\lambda)z + \beta = 0$. With

$$\eta = \frac{4}{(\sqrt{L}+\sqrt{\mu})^2}, \qquad \beta = \left(\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}\right)^2,$$

both roots are complex conjugate with modulus $\sqrt{\beta} = 1 - 2/(\sqrt{\kappa}+1) = 1 - O(1/\sqrt{\kappa})$ uniformly in λ, giving the claimed $\sqrt{\kappa}$ acceleration. $\square$

Proposition 0.140 (Decoupled weight decay $\neq L_2$ regularization for Adam)**.** *For SGD, augmenting the loss by $\frac{\lambda}{2}\|\theta\|^2$ adds $\lambda\theta$ to the gradient, producing the multiplicative shrinkage $\theta \mapsto (1-\eta\lambda)\theta$; the two formulations coincide. For Adam, the L_2 term enters g_t and is rescaled by $1/(\sqrt{\hat{v}_t}+\varepsilon)$: parameters with large historical gradients are decayed less, parameters with small historical gradients decayed more. AdamW restores the intended uniform shrinkage by applying $\lambda\theta_t$ outside the adaptive rescaling.*

Code sketch

```
def adam_step(theta, g, m, v, t, eta=1e-3, b1=0.9, b2=0.999, eps=1e-8):
    m = b1*m + (1 - b1)*g
    v = b2*v + (1 - b2)*g*g
    m_hat = m / (1 - b1**t)
    v_hat = v / (1 - b2**t)
    theta = theta - eta * m_hat / (np.sqrt(v_hat) + eps)
    return theta, m, v
```

AdamW differs by a single line: `theta = theta - eta*(m_hat/(np.sqrt(v_hat)+eps) + lam*theta)`.

Connection to LLMs

AdamW is the universal pre-training optimizer for transformers (GPT-2/3/4 family, Llama, Mistral, Gemma). Two facts from the analysis above explain why. (i) Per-coordinate rescaling matters: embedding rows, LayerNorm scales, and dense projection weights see gradients differing in magnitude by 3–6 orders of magnitude; only adaptive optimizers train them all under one global learning rate. (ii) Decoupled weight decay is essential for generalization at scale — with vanilla L_2-Adam, infrequently-updated parameters such as rare-token embeddings would be barely regularized while high-gradient parameters would be over-shrunk. Bias correction matters most in the first $\sim 10^1$–10^3 steps; the $1/(1-\beta_1)$ effective horizon explains why warmup of a few hundred steps is standard. Concrete AdamW configuration ($\lambda = 0.1$, $\beta_2 = 0.95$, gradient clipping, warmup-then-decay schedule) is taken up in Chapter 27.

Block D — Neural Networks

Chapter 15: MLPs as compositional functions; universal approximation

Motivation

A linear map $x \mapsto Wx + b$ (Chapter 5) can only carve $\mathbb{R}^n$ into half-spaces and stretch them affinely. The continuous functions on a compact set $K \subset \mathbb{R}^n$ (Chapter 4) form a vastly richer space. To bridge the gap we interleave linear maps with a fixed pointwise nonlinearity. The resulting object — a *multilayer perceptron* (MLP) — is dense in $C(K)$. This is the engine inside every transformer's feed-forward block.

Definitions

Definition 0.141 (Multilayer perceptron)**.** *Let $L \geq 1$ and widths $d_0, d_1, \ldots, d_L \in \mathbb{N}$. A* multilayer perceptron *of depth L with activation $\sigma : \mathbb{R} \to \mathbb{R}$ (applied componentwise) is the function $f : \mathbb{R}^{d_0} \to \mathbb{R}^{d_L}$ defined by*

$$\begin{aligned} h^{(0)} &= x, \\ h^{(\ell)} &= \sigma\big(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}\big), \quad 1 \leq \ell \leq L-1, \\ f(x) &= W^{(L)} h^{(L-1)} + b^{(L)}, \end{aligned}$$

where $W^{(\ell)} \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$ and $b^{(\ell)} \in \mathbb{R}^{d_\ell}$. We call $\max_{0 \leq \ell \leq L} d_\ell$ the width *and L the* depth.

Definition 0.142 (Universal approximator)**.** *A class* $\mathcal{F} \subset C(K,\mathbb{R})$ *is* dense *in* $C(K)$ *— equivalently a* universal approximator *— if*

$$\forall f \in C(K)\ \forall \varepsilon > 0\ \exists g \in \mathcal{F} : \sup_{x \in K} |f(x) - g(x)| < \varepsilon.$$

Definition 0.143 (Sigmoidal, discriminatory)**.** *A measurable* $\sigma : \mathbb{R} \to \mathbb{R}$ *is* sigmoidal *if* $\sigma(t) \to 1$ *as* $t \to +\infty$ *and* $\sigma(t) \to 0$ *as* $t \to -\infty$*. It is* discriminatory *on* $K = [0,1]^n$ *if for every finite signed regular Borel measure* μ *on* K*,*

$$\int_K \sigma(w^\top x + b)\, d\mu(x) = 0\ \ \forall\, w \in \mathbb{R}^n,\ b \in \mathbb{R} \implies \mu = 0.$$

Theorems and proofs

Theorem 0.144 (Cybenko 1989; Hornik 1989)**.** *Let* $\sigma : \mathbb{R} \to \mathbb{R}$ *be a continuous sigmoidal function and let* $K \subset \mathbb{R}^n$ *be compact. The class*

$$\mathcal{F}_\sigma = \Big\{ g(x) = \sum_{j=1}^{N} \alpha_j\, \sigma(w_j^\top x + b_j) : N \in \mathbb{N},\ \alpha_j, b_j \in \mathbb{R},\ w_j \in \mathbb{R}^n \Big\}$$

is dense in $C(K)$*.*

Proof sketch (Hahn–Banach + Riesz). Suppose, for contradiction, $\mathcal{F}_\sigma$ is *not* dense in $C(K)$. Then $S := \overline{\mathcal{F}_\sigma}$ is a proper closed subspace of the Banach space $\big(C(K), \|\cdot\|_\infty\big)$. Pick $f_0 \in C(K) \setminus S$. By the Hahn–Banach theorem there exists a continuous linear functional $\Lambda : C(K) \to \mathbb{R}$ with

$$\Lambda \not\equiv 0, \qquad \Lambda(g) = 0 \text{ for all } g \in S.$$

The Riesz–Markov representation theorem identifies $C(K)^*$ with the space of finite signed regular Borel measures on K: there exists a nonzero such measure μ with

$$\Lambda(g) = \int_K g(x)\, d\mu(x), \qquad g \in C(K).$$

Each ridge function $x \mapsto \sigma(w^\top x + b)$ lies in $\mathcal{F}_\sigma \subset S$, hence

$$\int_K \sigma(w^\top x + b)\, d\mu(x) = 0 \qquad \forall w \in \mathbb{R}^n,\ b \in \mathbb{R}.$$

Cybenko's lemma asserts that every continuous sigmoidal σ is discriminatory. (Sketch: as $\lambda \to \infty$, $\sigma(\lambda(w^\top x + b) + \varphi)$ converges pointwise (and boundedly) to a step function of the half-space $\{w^\top x + b > 0\}$; the dominated-convergence theorem transfers vanishing of the ridge integrals to vanishing of μ on every half-space, and finite signed measures are determined by their values on half-spaces, e.g. via the Fourier transform.) Therefore $\mu = 0$, contradicting $\Lambda \not\equiv 0$. Hence $\mathcal{F}_\sigma$ is dense. $\square$

Proposition 0.145 (Linear-only networks collapse)**.** *If* $\sigma = \mathrm{id}$*, every depth-L MLP is itself a single affine map. In particular, the class is not dense in* $C(K)$ *whenever* K *contains a nontrivial open set.*

Proof. By induction on ℓ, $h^{(\ell)} = W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}$. Unrolling,

$$f(x) = W'x + b', \qquad W' = W^{(L)} W^{(L-1)} \cdots W^{(1)}, \qquad b' = \sum_{\ell=1}^{L} \Big(\prod_{k=L}^{\ell+1} W^{(k)} \Big) b^{(\ell)},$$

where the empty product (when $\ell = L$) is the identity. Thus f is affine. An affine f cannot uniformly approximate, e.g., $x \mapsto \sin(2\pi x)$ on $[-1, 1]$ to error $< 1/4$, since the best affine approximant on a symmetric interval is 0 and $\|\sin(2\pi\cdot)\|_\infty = 1$. $\square$

Remark 0.146. Proposition 0.145 shows the nonlinearity σ is not cosmetic: *depth without nonlinearity buys nothing.* UAT therefore relies on σ being genuinely nonlinear.

Theorem 0.147 (Depth separation, Telgarsky 2016)**.** *For every $k \in \mathbb{N}$ there is a function $f_k : [0, 1] \to [0, 1]$ realizable* exactly *by a ReLU network of depth $O(k^3)$ and width $O(1)$ such that any ReLU network of depth at most k that approximates f_k in $L^1([0, 1])$ to error $< 1/32$ must have width $\geq 2^k$.*

Proof sketch. Let $\Delta : [0, 1] \to [0, 1]$ be the triangular sawtooth $\Delta(x) = 2x$ for $x \in [0, 1/2]$, $\Delta(x) = 2(1 - x)$ for $x \in [1/2, 1]$, and define $f_k = \Delta^{\circ k}$. Each composition doubles the number of monotone pieces, so f_k has 2^k such pieces and L^1-distance $1/4$ from any function with $\leq 2^{k-1}$ pieces. A ReLU network of width w and depth d realizes a continuous piecewise-linear function with at most $(2w)^d$ pieces (each ReLU at most doubles the count). Forcing $(2w)^d \geq 2^{k-1}$ at $d = k$ yields $w \geq 2^{(k-1)/k-1} \cdot 2^{k/k}$; a careful counting argument tightens this to the stated 2^k. The composition $\Delta^{\circ k}$ on the other hand is realised by stacking k copies of the constant-width gadget that implements Δ. $\square$

Remark 0.148. Theorem 0.144 says shallow nets *can* represent everything; Theorem 0.147 says compositionality (depth) is *exponentially more parameter-efficient* for hierarchical targets. Modern deep learning lives in the gap between these two statements.

Code sketch

The companion notebook `cells.json`: (i) hand-builds a numpy MLP with tanh activation; (ii) fits a 32-hidden-unit MLP to $\sin(2\pi x)$ on $[-1, 1]$ via least-squares on hidden features and verifies $\sup_x |f(x) - \hat{f}(x)| < 0.05$; (iii) compares width-32-depth-1 against width-8-depth-3 at matched parameter budget and observes lower error from the deeper net; (iv) numerically confirms the linear-only collapse $f(x) = W'x + b'$ predicted by Proposition 0.145.

Connection to LLMs

Every transformer block (Chapter 23, forthcoming) contains a position-wise feed-forward module

$$\mathrm{FFN}(x) = W_2\, \sigma(W_1 x + b_1) + b_2,$$

i.e. a single-hidden-layer MLP applied independently to each token. This is exactly the universal approximator of Theorem 0.144: attention mixes information *across* positions, while the FFN does the nonlinear pointwise computation that — by Cybenko — can in principle realise any continuous transformation of the embedding. LLMs make W_1 wide ($4d$ or $8d$ inner dimension) precisely to exploit Theorem 0.144's expressive guarantee, while stacking L such blocks exploits Theorem 0.147's exponential depth advantage.

Chapter 16: Activation functions: ReLU/GELU/softmax with derivatives

Motivation

A multi-layer perceptron (Chapter 15) without a nonlinear activation collapses into a single affine map: $W_2(W_1x+b_1)+b_2 = (W_2W_1)x+(W_2b_1+b_2)$. Universal approximation requires a pointwise nonlinearity ϕ between affine layers. The choice of ϕ controls the gradient signal that backpropagation (Chapter 3) is allowed to push through the network, the geometry of hidden activations, and the numerical conditioning of training. We derive the five activations that dominate modern deep learning—sigmoid, tanh, ReLU, GELU, and softmax—computing each derivative or Jacobian from first principles and analyzing saturation.

Definitions

Definition 0.149 (Sigmoid). $\sigma : \mathbb{R} \to (0,1)$, $\sigma(x) = \dfrac{1}{1+e^{-x}}$.

Definition 0.150 (Hyperbolic tangent). $\tanh(x) = \dfrac{e^x - e^{-x}}{e^x + e^{-x}} = 2\sigma(2x) - 1$.

Definition 0.151 (ReLU). $\mathrm{ReLU}(x) = \max(0,x)$. *Differentiable except at* $x = 0$*; the Clarke subdifferential at* 0 *is* $[0,1]$.

Definition 0.152 (GELU). $\mathrm{GELU}(x) = x\,\Phi(x)$, *where* $\Phi(x) = \frac{1}{2}\big(1+\mathrm{erf}(x/\sqrt{2})\big)$ *is the standard-normal CDF. The tanh-approximation is*

$$\mathrm{GELU}(x) \;\approx\; \tfrac{1}{2}x\big(1+\tanh(\sqrt{2/\pi}\,(x+0.044715\,x^3))\big).$$

Definition 0.153 (Softmax). $\mathrm{softmax} : \mathbb{R}^n \to \Delta^{n-1}$, $\mathrm{softmax}(z)_i = \dfrac{e^{z_i}}{\sum_{k=1}^n e^{z_k}}$.

Theorems and Proofs

Theorem 0.154 (Sigmoid derivative). $\sigma'(x) = \sigma(x)\big(1-\sigma(x)\big)$.

Proof. Write $\sigma(x) = (1+e^{-x})^{-1}$. By the chain rule, $\sigma'(x) = -(1+e^{-x})^{-2}\cdot(-e^{-x}) = \dfrac{e^{-x}}{(1+e^{-x})^2}$. Factor:

$$\frac{e^{-x}}{(1+e^{-x})^2} = \frac{1}{1+e^{-x}}\cdot\frac{e^{-x}}{1+e^{-x}} = \sigma(x)\big(1-\sigma(x)\big),$$

using $\frac{e^{-x}}{1+e^{-x}} = 1-\sigma(x)$. □

Theorem 0.155 (Tanh derivative). $\tanh'(x) = 1-\tanh^2(x)$.

Proof. From $\tanh(x) = 2\sigma(2x)-1$ and Theorem 0.154, $\tanh'(x) = 4\sigma(2x)(1-\sigma(2x))$. Substituting $\sigma(2x) = (\tanh x + 1)/2$,

$$4\cdot\tfrac{\tanh x+1}{2}\cdot\tfrac{1-\tanh x}{2} = (1+\tanh x)(1-\tanh x) = 1-\tanh^2 x. \qquad \square$$

Theorem 0.156 (ReLU subdifferential). *For* $x \neq 0$, $\mathrm{ReLU}'(x) = \mathbf{1}_{x>0}$. *At* $x = 0$, $\partial\,\mathrm{ReLU}(0) = [0,1]$.

Proof. For $x > 0$, $\mathrm{ReLU}(x) = x$ has derivative 1. For $x < 0$, derivative 0. At $x = 0$ the left derivative is 0 and the right derivative is 1; the convex hull of these one-sided limits, $[0, 1]$, defines the Clarke subdifferential. PyTorch and JAX conventionally pick 0. $\square$

Theorem 0.157 (GELU derivative)**.** $\mathrm{GELU}'(x) = \Phi(x) + x\,\phi(x)$*, where* $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$*.*

Proof. By the product rule, $\frac{d}{dx}\big[x\Phi(x)\big] = \Phi(x) + x\,\Phi'(x)$. The fundamental theorem of calculus gives $\Phi'(x) = \phi(x)$. $\square$

Theorem 0.158 (Softmax Jacobian)**.** *Let* $s = \mathrm{softmax}(z)$ *and* $S = \sum_k e^{z_k}$*. Then*

$$\frac{\partial s_i}{\partial z_j} = s_i(\delta_{ij} - s_j), \qquad J(z) = \mathrm{diag}(s) - ss^\top.$$

Proof. By the quotient rule,

$$\frac{\partial s_i}{\partial z_j} = \frac{(\partial e^{z_i}/\partial z_j)\,S - e^{z_i}\,(\partial S/\partial z_j)}{S^2} = \frac{\delta_{ij}\,e^{z_i}S - e^{z_i}e^{z_j}}{S^2} = \delta_{ij}\,s_i - s_i s_j. \qquad \square$$

The matrix $J = \mathrm{diag}(s) - ss^\top$ is symmetric PSD with $J\mathbf{1} = 0$, reflecting softmax's translation invariance.

Proposition 0.159 (Saturation)**.** $\sigma'(x), \tanh'(x) \to 0$ *as* $|x| \to \infty$*, so backpropagated gradients* $\prod_\ell \phi'(z_\ell)$ *shrink geometrically in deep networks (vanishing gradients). ReLU avoids saturation for* $x > 0$ *but a unit with* $z \le 0$ *on every training input is permanently dead. GELU is smooth;* $\mathrm{GELU}'(x) \to 1$ *as* $x \to +\infty$ *(since* $\Phi \to 1$*,* $x\phi \to 0$*) and* $\to 0$ *as* $x \to -\infty$*, combining ReLU-like sparsity with smoothness.*

Theorem 0.160 (Temperature limits)**.** *Let* $T > 0$ *and assume* $i^* = \arg\max_i z_i$ *is unique. Then*

$$\lim_{T\to 0^+} \mathrm{softmax}(z/T)_i = \mathbf{1}_{i=i^*}, \qquad \lim_{T\to\infty} \mathrm{softmax}(z/T)_i = \tfrac{1}{n}.$$

Proof. Write $\mathrm{softmax}(z/T)_i = \dfrac{e^{(z_i - z_{i^*})/T}}{\sum_k e^{(z_k - z_{i^*})/T}}$. As $T \to 0^+$, every term with $k \ne i^*$ has exponent $\to -\infty$, so $\to 0$, while $k = i^*$ contributes 1, yielding $\mathbf{1}_{i=i^*}$. As $T \to \infty$, every exponent $\to 0$, so each term $\to 1$ and the ratio $\to 1/n$. $\square$

Proposition 0.161 (Numerical stability)**.** $\mathrm{softmax}(z) = \mathrm{softmax}(z - c\mathbf{1})$ *for any* c*. With* $c = \max_i z_i$*, all exponents are* ≤ 0 *and overflow is avoided.*

Proof. $\dfrac{e^{z_i - c}}{\sum_k e^{z_k - c}} = \dfrac{e^{-c}e^{z_i}}{e^{-c}\sum_k e^{z_k}} = \mathrm{softmax}(z)_i.$ $\square$

Corollary 0.162 (Log-sum-exp)**.** $\log \sum_j e^{z_j} = \max_j z_j + \log \sum_j e^{z_j - \max_k z_k}$.

Code Sketch

The companion notebook plots all five activations on $[-4, 4]$ and tabulates them at $x \in \{-2, -1, 0, 1, 2\}$; verifies each analytic derivative against centered finite differences; implements `softmax_jacobian(z) = np.diag(s) - np.outer(s,s)` and confirms it matches a numerical Jacobian for $z = (1, 0.5, -1, 2)$; sweeps $T \in \{0.1, 1, 5, 100\}$ to visualize the temperature limits; and shows that naive softmax of $(1000, 1001, 1002)$ overflows while the stabilized version returns $(0.0900, 0.2447, 0.6652)$.

Connection to LLMs

GELU is the standard feedforward activation in GPT-2/3 and BERT, applied elementwise to the $4d_{\text{model}}$ hidden state of every Transformer MLP block (Chapter 15). Llama-family models replace it with SwiGLU, a gated variant covered in Chapter 17. Softmax appears in two distinct loci of an LLM:

1. *Attention* (Chapter 21): row-wise $\text{softmax}(qK^\top/\sqrt{d_k})$ converts compatibility scores into a distribution over keys.
2. *Output head* (Chapter 25): final vocabulary logits are passed through softmax to produce $p(x_{t+1} \mid x_{\le t})$. The temperature limits (Theorem above) are the standard sampling knob: $T \to 0$ recovers greedy decoding, $T \to \infty$ uniform random sampling.

Both invocations use the stabilized form of Proposition above in production kernels (FlashAttention, fused log-softmax + cross-entropy), making the "subtract the max" identity load-bearing for trillion-parameter training.

Chapter 17: Loss functions: MSE, cross-entropy; gradients from first principles

Motivation

A neural network is a parametric map $f_\theta : \mathcal{X} \to \mathcal{Y}$. Training requires a scalar "badness" score whose gradient with respect to θ tells us how to nudge weights. That scalar is the *loss function.* Two losses dominate modern deep learning: **mean squared error** (MSE) for regression, and **categorical cross-entropy** (CE) for classification, including next-token prediction in every language model. Neither is arbitrary: each is the negative log-likelihood of a generative model (Ch. 12), and in both cases the gradient with respect to the network's pre-activation output collapses to $\hat{p} - y$, an identity that makes backpropagation through deep stacks tractable.

Definitions

Definition 0.163 (MSE loss)**.** *For scalar prediction $\hat{y}$ and target y,*

$$\ell_{\text{MSE}}(y, \hat{y}) \;=\; \tfrac{1}{2}(y - \hat{y})^2.$$

The factor $\frac{1}{2}$ is conventional; it cancels in the gradient.

Definition 0.164 (Categorical cross-entropy)**.** *For a one-hot label $y \in \{0,1\}^K$ with $\sum_k y_k = 1$ and a predicted distribution $\hat{p} \in \Delta^{K-1}$,*

$$\ell_{\text{CE}}(y, \hat{p}) \;=\; -\sum_{k=1}^{K} y_k \log \hat{p}_k.$$

This is $H(y, \hat{p})$ from Ch. 11 evaluated on a single example.

Definition 0.165 (Cross-entropy with logits)**.** *For logits $z \in \mathbb{R}^K$ and true class $c \in \{1, \ldots, K\}$,*

$$\ell(z, c) \;=\; -\log \text{softmax}(z)_c \;=\; -z_c + \log \sum_{j=1}^{K} e^{z_j}.$$

The right-hand log-sum-exp form is the only numerically stable evaluation route.

Definition 0.166 (Binary cross-entropy). *For $K = 2$ with logit z, $\hat{p} = \sigma(z)$, and $y \in \{0, 1\}$,*

$$\ell(z, y) \;=\; -y \log \sigma(z) - (1 - y) \log(1 - \sigma(z)).$$

Theorems and proofs

Theorem 0.167 (MSE gradient). $\nabla_{\hat{y}}\, \ell_{\mathrm{MSE}}(y, \hat{y}) = \hat{y} - y$.

Proof. $\frac{d}{d\hat{y}}\big[\frac{1}{2}(y - \hat{y})^2\big] = -(y - \hat{y}) = \hat{y} - y$. □

Theorem 0.168 (MSE = MLE under Gaussian noise). *If $y = f_\theta(x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, then maximum likelihood estimation is equivalent to MSE minimization.*

Proof. The Gaussian density gives

$$-\log p_\theta(y \mid x) \;=\; \tfrac{1}{2}\log(2\pi\sigma^2) + \frac{(y - f_\theta(x))^2}{2\sigma^2} \;=\; \frac{1}{2\sigma^2}(y - f_\theta(x))^2 + C,$$

where C is independent of θ. Summing across an i.i.d. dataset and dropping the additive constant and the positive multiplier $1/\sigma^2$ yields $\sum_n \frac{1}{2}(y_n - f_\theta(x_n))^2$, the MSE objective. The argmin in θ coincides with the argmax of the log-likelihood. □

Theorem 0.169 (Categorical CE = MLE under softmax). *Let $p_\theta(y = c \mid x) = \mathrm{softmax}(z_\theta(x))_c$. Then $-\log p_\theta(y = c \mid x) = \ell(z, c)$, so MLE summed over an i.i.d. dataset equals categorical CE minimization.*

Proof. Direct from the definition of $\ell(z, c)$ and the identity $\arg\max \log L = \arg\min H_{\mathrm{cross}}$ proved in Ch. 12. □

Theorem 0.170 (Softmax + CE gradient). *For $\ell(z, c) = -z_c + \log \sum_j e^{z_j}$,*

$$\frac{\partial \ell}{\partial z_j} \;=\; \mathrm{softmax}(z)_j - \mathbf{1}_{[j=c]} \;=\; \hat{p}_j - y_j.$$

Proof. $\partial(-z_c)/\partial z_j = -\mathbf{1}_{[j=c]}$. For the log-sum-exp,

$$\frac{\partial}{\partial z_j} \log \sum_k e^{z_k} \;=\; \frac{e^{z_j}}{\sum_k e^{z_k}} \;=\; \mathrm{softmax}(z)_j.$$

Adding gives $\hat{p}_j - \mathbf{1}_{[j=c]}$. The cancellation occurs because the $\sum_k e^{z_k}$ in softmax's denominator is the *same* normalizer that appears inside ℓ; the matrix-vector product implicit in the softmax Jacobian (Ch. 16) collapses to a vector subtraction. One subtraction per output unit replaces a $K \times K$ matvec. □

Theorem 0.171 (Binary CE + sigmoid gradient). *For $\ell(z, y) = -y \log \sigma(z) - (1 - y) \log(1 - \sigma(z))$, $\partial \ell / \partial z = \sigma(z) - y$.*

Proof. Use $\log \sigma(z) = -\log(1 + e^{-z})$ and $\log(1 - \sigma(z)) = -z - \log(1 + e^{-z})$, so $\ell = (1 - y)\, z + \log(1 + e^{-z})$. Differentiating,

$$\frac{\partial \ell}{\partial z} \;=\; (1 - y) - \frac{e^{-z}}{1 + e^{-z}} \;=\; (1 - y) - (1 - \sigma(z)) \;=\; \sigma(z) - y. \qquad \square$$

Theorem 0.172 (CE minimum at $\hat{p} = y$). *Treating $y, \hat{p} \in \Delta^{K-1}$ as distributions, $H(y, \hat{p})$ is minimized over $\hat{p}$ at $\hat{p} = y$.*

Proof. $H(y, \hat{p}) - H(y, y) = -\sum_k y_k \log(\hat{p}_k / y_k) = D_{\mathrm{KL}}(y \,\|\, \hat{p}) \geq 0$ by Gibbs' inequality (Ch. 11), with equality iff $\hat{p} = y$. □

Code sketch

```
def softmax_ce_with_logits(z, c):
    z = z - z.max()
    lse = np.log(np.exp(z).sum())
    loss = -z[c] + lse
    p = np.exp(z - lse)
    grad = p.copy(); grad[c] -= 1.0      # p - y
    return loss, grad
```

The forward and backward share the cached softmax; the backward is a single in-place subtraction.

Connection to LLMs

A causal language model (Ch. 25) emits, at every position t, logits $z_t \in \mathbb{R}^V$ over a vocabulary of size V. The training objective on a sequence $x_1, \ldots, x_T$ is

$$\mathcal{L}(\theta) \;=\; \sum_{t=1}^{T} -\log p_\theta(x_t \mid x_{<t}) \;=\; \sum_{t=1}^{T} \ell(z_t, x_t),$$

i.e. cross-entropy with logits, summed across positions. By Theorem 4, the gradient at the output layer is $\hat{p}_t - y_t$ at every position — a single subtraction per token per vocabulary entry. This signal flows backwards through the unembedding, the transformer blocks (Ch. 27 forward), and into the token embeddings. With $V \approx 10^5$, evaluating the softmax Jacobian explicitly would be prohibitive; the $\hat{p} - y$ collapse is what makes large-scale language modeling computationally feasible.

Remark 0.173. The pattern "output-side gradient = prediction − target" generalizes well beyond softmax-CE and sigmoid-BCE: it is the canonical gradient for any *matched* pairing of an exponential-family likelihood with its natural-parameter activation (a fact we will revisit when discussing conjugate priors and generalized linear models).

Chapter 18: Backpropagation: chain rule applied; reverse-mode AD as a graph algorithm

Motivation

Chapter 17 ended with a clean fact: for softmax + cross-entropy, the gradient of the loss with respect to the pre-softmax logits is $\hat{p} - y$. That gives us the *output-layer* gradient. To train an actual network we need the gradient of the loss with respect to *every* parameter. Symbolic differentiation produces expressions whose size explodes exponentially with depth; finite differences over P parameters cost $O(P)$ forward passes, prohibitive when $P \approx 10^{11}$. Backpropagation solves both: it is the multivariate chain rule (Ch. 4) applied along a DAG in the right order.

Computational graphs and AD

Definition 0.174 (Computational graph). *A* computational graph *is a DAG whose nodes are intermediate values $v_i \in \mathbb{R}^{d_i}$ and whose edges are elementary operations. Source nodes are inputs; sink nodes are outputs. A* forward pass *computes all v_i in topological order given the inputs.*

Definition 0.175 (Forward-mode AD / JVP)**.** *Given a tangent $\dot{x}$, forward-mode AD propagates $\dot{v}_i = \sum_{j\in\text{parents}(i)}(\partial v_i/\partial v_j)\,\dot{v}_j$ in topological order. One sweep evaluates $J\dot{x}$. Cost: one sweep per* input *dimension.*

Definition 0.176 (Reverse-mode AD / VJP)**.** *Given a cotangent $\bar{y}$ at the output, reverse-mode AD propagates* adjoints *$\bar{v}_i$ in reverse topological order:*

$$\bar{v}_j \;=\; \sum_{i\in\text{children}(j)} \Big(\frac{\partial v_i}{\partial v_j}\Big)^T \bar{v}_i.$$

One sweep evaluates $J^T\bar{y}$. Cost: one sweep per output *dimension. For a scalar loss L, a single backward pass yields $\nabla_x L$; the adjoint $\bar{v}_i := \partial L/\partial v_i$ is the central object of backprop.*

The backprop theorem

Theorem 0.177 (Backprop = reverse-mode AD on a feedforward net)**.** *Let $L(x) = \ell(h^{(L)})$ where $h^{(\ell)} = f^{(\ell)}(h^{(\ell-1)})$, $h^{(0)} = x$, and each $f^{(\ell)}$ is differentiable with Jacobian $J^{(\ell)} := \partial f^{(\ell)}/\partial h^{(\ell-1)}$. Define adjoints $\bar{h}^{(\ell)} := \partial L/\partial h^{(\ell)}$. Then*

$$\bar{h}^{(L)} \;=\; \nabla \ell(h^{(L)}), \qquad \bar{h}^{(\ell-1)} \;=\; \big(J^{(\ell)}\big)^T \bar{h}^{(\ell)}.$$

Proof. By the multivariate chain rule (Ch. 4), for any ℓ,

$$\frac{\partial L}{\partial h^{(\ell-1)}} \;=\; \frac{\partial L}{\partial h^{(\ell)}} \cdot \frac{\partial h^{(\ell)}}{\partial h^{(\ell-1)}} \;=\; \bar{h}^{(\ell)}\, J^{(\ell)},$$

which transposed gives the adjoint recurrence. The base case is the gradient of ℓ at $h^{(L)}$. Induction over $\ell = L, L-1, \ldots, 1$. □

Remark 0.178. For Ch. 17's softmax+cross-entropy block, $\bar{h}^{(L)} = \hat{p} - y$ is the base case; the recurrence does the rest.

VJPs for an MLP layer

Proposition 0.179 (MLP layer VJPs)**.** *For $z^{(\ell)} = W^{(\ell)}h^{(\ell-1)} + b^{(\ell)}$, $h^{(\ell)} = \sigma(z^{(\ell)})$ with elementwise σ,*

$$\bar{z}^{(\ell)} \;=\; \bar{h}^{(\ell)} \odot \sigma'(z^{(\ell)}), \quad \bar{W}^{(\ell)} \;=\; \bar{z}^{(\ell)}(h^{(\ell-1)})^T, \quad \bar{b}^{(\ell)} \;=\; \bar{z}^{(\ell)}, \quad \bar{h}^{(\ell-1)} \;=\; (W^{(\ell)})^T \bar{z}^{(\ell)}.$$

Proof. The chain rule gives $\bar{z}_i^{(\ell)} = \sum_k \bar{h}_k^{(\ell)}\,\partial h_k^{(\ell)}/\partial z_i^{(\ell)} = \bar{h}_i^{(\ell)}\,\sigma'(z_i^{(\ell)})$ since σ is elementwise. For W, $z_i = \sum_j W_{ij} h_j^{(\ell-1)}$ so $\partial z_i/\partial W_{kj} = \delta_{ik} h_j^{(\ell-1)}$, hence $\bar{W}_{kj} = \bar{z}_k h_j^{(\ell-1)}$, an outer product. The bias is $\partial z_i/\partial b_j = \delta_{ij}$. For $h^{(\ell-1)}$, $\partial z_i/\partial h_j^{(\ell-1)} = W_{ij}$, giving $\bar{h}^{(\ell-1)} = W^T \bar{z}^{(\ell)}$. □

Cost and memory

Theorem 0.180 (Baur–Strassen)**.** *For a function defined by K elementary operations with bounded fan-out, the gradient can be computed in arithmetic cost at most $\sim 5K$ — i.e. the backward pass costs only a small constant times the forward pass,* independent of the number of parameters *P.*

This is the magic. We pay a constant overhead, not a P-fold one, to differentiate with respect to all P parameters simultaneously. Forward-mode AD does the opposite: $O(n)$ for n inputs, $O(1)$ in outputs.

Remark 0.181 (Memory). The recurrence $\bar{h}^{(\ell-1)} = (J^{(\ell)})^T \bar{h}^{(\ell)}$ requires evaluating $J^{(\ell)}$, which typically depends on the forward activation $h^{(\ell-1)}$ (e.g. via $\sigma'(z^{(\ell)})$). Hence reverse-mode AD must *cache every intermediate activation* produced during the forward pass, giving memory cost $O(L)$ in depth times batch and width. This is exactly the constraint that motivates *gradient checkpointing* (Ch. 27): drop activations and recompute them on backward, trading compute for memory.

Code sketch

Example 0.182 (Two-line backward for an MLP layer).
```
def linear_vjp(W, h_in, dz):
    return dz @ h_in.T, dz, W.T @ dz   # dW, db, dh_in

def tanh_vjp(z, dh):
    return dh * (1.0 - np.tanh(z)**2)
```
The full backward pass over L layers is L applications of these two VJPs in reverse, threading the adjoint $\bar{h}^{(\ell)}$ through.

Connection to LLMs

A modern transformer has $L \in [12, 96]$ blocks, each composed of attention and MLP sub-layers. One training step is: (i) forward pass through all L blocks, caching activations; (ii) reverse-mode AD computing $\nabla_\theta L$ for every parameter (often $\theta \in \mathbb{R}^{10^{11}}$) in time $\leq 5\times$ the forward pass — Baur–Strassen in action; (iii) optimizer step. The per-token activation memory is the dominant scaling constraint and is exactly why frameworks ship gradient checkpointing, FlashAttention recomputation (Ch. 23), and ZeRO-style sharding by default. Every parameter update in a 175B model is the adjoint recurrence above, executed billions of times.

Block E — Sequence Models and Attention

Chapter 19: Embeddings: token to vector; lookup as a linear map; weight tying

Motivation

A transformer language model never sees the symbol *cat*. It sees an integer index $v \in \{0, 1, \ldots, V-1\}$ which is then mapped to a vector in $\mathbb{R}^d$. That vector is the only representation downstream layers ever touch: attention, MLPs, normalizations, residuals — everything is linear algebra on $\mathbb{R}^d$. The gateway between the discrete vocabulary $\mathcal{V}$ (Chapter 1) and the continuous geometry of $\mathbb{R}^d$ is the *embedding matrix*. We will show that this gateway is not a fancy table lookup at all: it is the linear map of Chapter 5 applied to a one-hot vector. The trick of *weight tying* — sharing parameters between the input embedding and the output projection — then falls out as a natural identification.

Definitions

Definition 0.183 (One-hot encoding). *For a finite vocabulary $\mathcal{V} = \{0, 1, \ldots, V-1\}$ and $v \in \mathcal{V}$, the* one-hot vector *is $e_v \in \{0,1\}^V$ with $e_v[i] = \mathbf{1}_{i=v}$. Equivalently, e_v is the v-th standard basis vector of $\mathbb{R}^V$.*

Definition 0.184 (Embedding matrix and lookup)**.** *An* embedding matrix *is a parameter* $E \in \mathbb{R}^{d\times V}$*. Its columns* $\{E_{:,0}, \ldots, E_{:,V-1}\}$ *are the* token embeddings*; the integer* d *is the* embedding dimension *(also written* d_{model}*). The* embedding lookup *is* $\text{emb}\colon \mathcal{V} \to \mathbb{R}^d$, $\text{emb}(v) := Ee_v$.

Definition 0.185 (Output projection / unembedding)**.** *An* output projection *is a parameter* $U \in \mathbb{R}^{V\times d}$ *taking a hidden state* $h \in \mathbb{R}^d$ *to a logit vector* $z := Uh \in \mathbb{R}^V$*. The probability over* $\mathcal{V}$ *is* $\hat{p} = \text{softmax}(z)$ *(Chapter 9, 17).*

Definition 0.186 (Weight tying)**.** *A model with embedding* $E \in \mathbb{R}^{d\times V}$ *and output projection* $U \in \mathbb{R}^{V\times d}$ *is* weight-tied *(Press & Wolf 2016; Inan et al. 2017) if* $U = E^\top$*. The two layers then share the same* dV *scalar parameters.*

Theorems and proofs

Theorem 0.187 (Lookup is a linear map)**.** *For every* $v \in \{0, \ldots, V-1\}$, $Ee_v = E_{:,v}$.

Proof. By the definition of matrix–vector multiplication (Chapter 5),

$$(Ee_v)_i \;=\; \sum_{j=0}^{V-1} E_{ij}\,(e_v)_j \;=\; \sum_{j=0}^{V-1} E_{ij}\,\mathbf{1}_{j=v} \;=\; E_{iv}$$

for every $i \in \{0, \ldots, d-1\}$. Stacking these scalars gives the column $E_{:,v}$, so $Ee_v = E_{:,v}$. □

Remark 0.188. Embedding-table lookup is therefore a linear map $\mathbb{R}^V \to \mathbb{R}^d$ in disguise. Implementations skip the matmul and slice the column; the *meaning* is the matrix product.

Theorem 0.189 (Weight-tying gradient identity)**.** *Let* $E \in \mathbb{R}^{d\times V}$ *and consider a model*

$$f(v, E) \;=\; U\,h(Ee_v) \;=\; E^\top h(Ee_v),$$

with tied $U = E^\top$*, where* $h\colon \mathbb{R}^d \to \mathbb{R}^d$ *is a sub-network with no further dependence on* E*. Let* $\mathcal{L}$ *be a scalar loss applied to* $z = E^\top h$*. Then*

$$\nabla_E \mathcal{L} \;=\; \underbrace{h\,(\nabla_z \mathcal{L})^\top}_{\textit{output-side}} \;+\; \underbrace{\big(J_h^\top E\,\nabla_z \mathcal{L}\big)\,e_v^\top}_{\textit{input-side}},$$

where $J_h \in \mathbb{R}^{d\times d}$ *is the Jacobian of* h *at* Ee_v*. The input-side contribution is rank-one; only column* v *is nonzero.*

Proof. Treat E as occurring in two roles: an output role $U = E^\top$ and an input role inside $h\circ(E\cdot e_v)$. By the multivariate chain rule (Chapter 18), the total gradient is the sum of the partial gradients with respect to each role.

Output role. $z_k = \sum_i E_{ik} h_i$, so $\partial z_k/\partial E_{ij} = \mathbf{1}_{k=j} h_i$. Hence

$$\frac{\partial \mathcal{L}}{\partial E_{ij}}\Big|_{\text{out}} \;=\; \sum_k \frac{\partial \mathcal{L}}{\partial z_k}\frac{\partial z_k}{\partial E_{ij}} \;=\; h_i\,\frac{\partial \mathcal{L}}{\partial z_j},$$

giving the outer product $h\,(\nabla_z \mathcal{L})^\top \in \mathbb{R}^{d\times V}$.

Input role. $x := Ee_v$ has $x_i = E_{iv}$, so $\partial x_i/\partial E_{ab} = \mathbf{1}_{a=i}\mathbf{1}_{b=v}$. By the chain rule,

$$\frac{\partial \mathcal{L}}{\partial E_{ab}}\Big|_{\text{in}} \;=\; \sum_i \frac{\partial \mathcal{L}}{\partial x_i}\frac{\partial x_i}{\partial E_{ab}} \;=\; \frac{\partial \mathcal{L}}{\partial x_a}\,\mathbf{1}_{b=v}.$$

Now $\nabla_x \mathcal{L} = J_h^\top \nabla_h \mathcal{L} = J_h^\top (E\,\nabla_z \mathcal{L})$, since $\nabla_h \mathcal{L} = U^\top \nabla_z \mathcal{L} = E\,\nabla_z \mathcal{L}$. Hence the input-side contribution is the rank-one matrix $g\,e_v^\top$ with $g := J_h^\top E\,\nabla_z \mathcal{L} \in \mathbb{R}^d$, whose only nonzero column sits in position v. Summing the two roles yields the claim. □

Remark 0.190 (Sparsity of the input gradient). The factor $e_v^\top$ kills every column except column v. This is why embedding tables are usually updated with *sparse* gradients: each minibatch only touches the columns of the tokens it actually contains. The output-side term, by contrast, hits all V columns through $h(\nabla_z \mathcal{L})^\top$.

Remark 0.191 (Distributional semantics, Mikolov et al. 2013). During training, two tokens u, v that appear in similar surrounding contexts produce similar gradients on their embeddings (the context enters via $\nabla_z \mathcal{L}$ and via h). Iterating SGD, $E_{:,u}$ and $E_{:,v}$ drift toward similar locations in $\mathbb{R}^d$, so cosine similarity of embeddings tracks distributional similarity in the corpus. This is an empirical claim, not a theorem; we illustrate it numerically in the code cells.

Theorem 0.192 (Dimensionality bound). *If $V > d$, no choice of $E \in \mathbb{R}^{d \times V}$ makes the columns $E_{:,0}, \ldots, E_{:,V-1}$ pairwise orthogonal and nonzero.*

Proof. Pairwise orthogonal nonzero vectors in any inner-product space are linearly independent: from $\sum_v c_v E_{:,v} = 0$, take the inner product with $E_{:,u}$ to get $c_u \|E_{:,u}\|^2 = 0$, hence $c_u = 0$. The columns would then be a linearly independent set of size V in $\mathbb{R}^d$. By Chapter 5 (invariance of dimension), every independent set in $\mathbb{R}^d$ has size $\leq d$, so $V \leq d$, contradicting $V > d$. □

In LLMs we always have $V \gg d$ (e.g. $V \approx 5 \cdot 10^4$ versus $d \approx 4096$), so embeddings cannot be mutually orthogonal — they form an *overcomplete* set, and "near-orthogonality" is the best one can ask.

Code sketch

We build a tiny embedding matrix and verify $Ee_v = E_{:,v}$ exactly. We then construct tied $U = E^\top$ and check that the logit for token v equals $\langle E_{:,v}, h \rangle$. A Mikolov-style skip-gram on a synthetic corpus shows that embeddings of co-occurring tokens develop higher cosine similarity than non-co-occurring ones. Finally, we verify Theorem 0.192 by attempting to orthogonalize $V = 8$ vectors in $\mathbb{R}^3$ via Gram–Schmidt and observing dimension exhaustion.

Connection to LLMs

Every transformer language model contains exactly the data structures of this chapter: an input embedding $E \in \mathbb{R}^{d \times V}$ and an output projection $U \in \mathbb{R}^{V \times d}$. The embedding dimension d is the *model dimension* d_{model}, which dominates parameter counts and FLOPs (Chapters 23–25). GPT-2 (Radford et al. 2019), Llama 1/2/3 (Touvron et al. 2023), and most open-weight LMs use weight tying $U = E^\top$, halving embedding parameters and empirically improving perplexity. GPT-3/4 use an LM-head initialized from $E^\top$ but trained separately. Once a token is embedded, position information is added (Chapter 20), and the residual stream begins (Chapter 22). The final logit projection Uh_{final} is the same matrix E that began the forward pass, viewed from a different side.

Chapter 20: RNN intuition; vanishing-gradient proof; why we need attention

Motivation

Feedforward networks (Chapters 17–19) are fixed-depth pipelines, but language, audio, and code are *sequences*. We need a model that consumes one token at a time, accumulates context, and emits one prediction per step. The Recurrent Neural Network (RNN) was the dominant answer from the late 1980s through 2017. We derive its dynamics, prove the *vanishing/exploding gradient theorem* (Pascanu, Mikolov, Bengio 2013) using the SVD of Chapter 6 and the chain rule of Chapter 18, and show that this failure mode—together with an information bottleneck—is the precise reason attention (Chapters 21–24) was invented.

Definitions

Definition 0.193 (Sequence model)**.** *A causal map $f : \mathcal{X}^T \to \mathcal{Y}^T$ such that $y_t = f_t(x_1, \ldots, x_t)$ for every t.*

Definition 0.194 (RNN cell)**.** *Fix hidden dimension d and parameters $W_h \in \mathbb{R}^{d\times d}$, $W_x \in \mathbb{R}^{d\times \dim x}$, $b \in \mathbb{R}^d$, and a pointwise nonlinearity σ. The recursion is*

$$h_t = \sigma(W_h h_{t-1} + W_x x_t + b), \qquad h_0 = 0,$$

with output $y_t = W_y h_t$.

Definition 0.195 (Backpropagation through time (BPTT))**.** *Unroll the recursion into an acyclic graph of depth T with the parameters (W_h, W_x, b) shared across every time step, then apply the backprop algorithm of Chapter 18 to that unrolled graph. The gradient of a shared weight is the sum of its per-step gradients.*

Theorems and proofs

Theorem 0.196 (Linear vanishing/exploding gradient; Pascanu et al. 2013)**.** *Let $h_t = W h_{t-1}$ with $W \in \mathbb{R}^{d\times d}$, and let L be a scalar loss depending on h_T. Then for any $t < T$,*

$$\frac{\partial L}{\partial h_t} = (W^\top)^{T-t}\,\frac{\partial L}{\partial h_T}, \qquad \left\|\tfrac{\partial L}{\partial h_t}\right\|_2 \le \sigma_{\max}(W)^{T-t}\left\|\tfrac{\partial L}{\partial h_T}\right\|_2,$$

with the upper bound achievable.

Proof. By the multivariable chain rule of Chapter 18, $\partial h_t/\partial h_{t-1} = W$, so $\partial L/\partial h_{t-1} = W^\top\, \partial L/\partial h_t$. Iterating $T-t$ times gives the claimed product. By Chapter 6, $W = U\Sigma V^\top$ with $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_d)$ and U, V orthogonal. Then $W^\top = V\Sigma U^\top$ and

$$(W^\top)^{T-t} = V\Sigma U^\top V\Sigma U^\top \cdots$$

which does *not* simplify in general unless we reuse the SVD of W^{T-t} instead: writing $W^{T-t} = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$, the singular values are bounded by $\sigma_{\max}(W)^{T-t}$ via submultiplicativity $\|AB\|_2 \le \|A\|_2\|B\|_2$ of the operator norm. Hence $\|(W^\top)^{T-t}\|_2 = \|W^{T-t}\|_2 \le \sigma_{\max}(W)^{T-t}$. Equality is achieved when W is normal (so W^{T-t} has singular values $|\lambda_i(W)|^{T-t}$) and $\partial L/\partial h_T$ is aligned with the top singular direction. □

Corollary 0.197 (Saturating nonlinearity)**.** *If σ has $|\sigma'| \le c$ pointwise (for* tanh*, $c = 1$), then with $D_t = \operatorname{diag}(\sigma'(\cdot))$ a diagonal Jacobian,*

$$\left\|\tfrac{\partial L}{\partial h_t}\right\|_2 \le c^{T-t}\,\|W\|_2^{T-t}\left\|\tfrac{\partial L}{\partial h_T}\right\|_2.$$

Proof. The Jacobian of one step is $D_t W$ with $\|D_t\|_2 \le c$. Apply submultiplicativity $T-t$ times. □

Remark 0.198. For tanh, saturation pushes $\|D_t\|_2$ *below* 1 in the asymptotic regions, so nonlinearity tends to *worsen* vanishing. This is the regime in which standard RNNs fail to learn dependencies beyond roughly $T - t \sim 20$–50 steps.

Theorem 0.199 (Information bottleneck of an RNN). *The hidden state $h_t \in \mathbb{R}^d$ is a deterministic function of $(x_1, \ldots, x_t)$, so by the data-processing inequality applied to the Markov chain $(x_1, \ldots, x_t) \to h_t \to \hat{x}_s$ for $s \leq t$, the mutual information satisfies $I(x_s; \hat{x}_s) \leq I(x_s; h_t) \leq d \cdot B$, where B is the per-coordinate bit precision. Once $t \log_2 |\mathcal{X}| \gg dB$, some information about earlier tokens is provably destroyed.*

Sketch. Apply the data-processing inequality (a standard result of information theory) to the deterministic Markov chain. The right inequality is the channel capacity of a d-dimensional vector at B bits per coordinate. □

Proposition 0.200 (Attention dissolves both bottlenecks — sketch). *Let $h_T = \sum_{s=1}^{T} \alpha_{Ts} V_s$ with values $V_s = V x_s$ and weights α_{Ts}. Then $\partial h_T / \partial V_s = \alpha_{Ts} I_d$ and $\partial L / \partial V_s = \alpha_{Ts} \, \partial L / \partial h_T$. This is a* single *linear step, independent of $T - s$, so no quantity is exponentiated in $T - s$. Each past token also has its own addressable slot, so the bottleneck of Theorem 0.199 is replaced by an $O(Td)$-sized cache. The full derivation appears in Chapter 21.*

Code sketch

The accompanying notebook (`cells.json`) implements an RNN cell with $d = 8$, $T = 50$, and seed 0. It empirically measures $\|\partial L / \partial h_t\|$ for spectral norms $\sigma_{\max}(W) \in \{0.5, 0.95, 1.0, 1.05, 1.5\}$ and plots log-norm versus $T - t$, recovering the linear (in log) decay/growth of Theorem 0.196. It then verifies $\sigma_{\max}(W^{T-t}) = \sigma_{\max}(W)^{T-t}$ (for normal W) via both SVD and power iteration, and shows that a one-shot attention average has gradient norm *independent* of position.

Connection to LLMs

The vanishing-gradient theorem is the historical reason transformer-based LLMs displaced RNNs. Bahdanau, Cho, and Bengio (2015) added attention to a seq2seq RNN to fix translation of long sentences; Vaswani et al. (2017, "Attention Is All You Need") removed recurrence entirely. Without an $O(1)$ gradient path, scaling to 10^4–10^6-token contexts is hopeless: a $\sigma_{\max} = 0.99$ RNN attenuates by e^{-100} over 10,000 steps. We formalize attention in Chapter 21, generalize to multi-head in Chapter 22, embed in the Transformer block in Chapter 23, and scale in Chapter 24.

Chapter 21: Scaled dot-product attention: derivation, softmax-temperature analysis

Motivation

Chapter 20 closed with a sharp negative result: a recurrent network propagates gradients between tokens j and i through $|i - j|$ multiplicative steps, so $\|\partial h_i / \partial h_j\|$ shrinks or grows geometrically. Long-range dependencies become statistically unreachable. We now construct a primitive that links every output position to every input position through a *single* matrix multiplication, so the gradient between any pair of tokens travels in $O(1)$ depth: **scaled dot-product attention** [1].

Definitions

Let $X \in \mathbb{R}^{T\times d_{\text{model}}}$ be a sequence of T token vectors, e.g. embeddings from Chapter 19. Fix projection matrices

$$W_Q, W_K \in \mathbb{R}^{d_{\text{model}}\times d_k}, \qquad W_V \in \mathbb{R}^{d_{\text{model}}\times d_v},$$

and define the **queries**, **keys**, and **values**

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V.$$

These are linear maps in the sense of Chapter 5.

Definition 0.201 (Scaled dot-product attention)**.**

$$\boxed{\mathrm{Attn}(Q,K,V) \;=\; \mathrm{softmax}\!\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.}$$

The softmax (Chapter 16) is applied row-wise. We write

$$A \;:=\; \mathrm{softmax}\!\left(\tfrac{QK^\top}{\sqrt{d_k}}\right) \in \mathbb{R}^{T\times T}$$

for the attention weights. *Each row $A_{i,:}$ is a probability vector: $A_{ij} \geq 0$ and $\sum_j A_{ij} = 1$.*

Definition 0.202 (Self- vs. cross-attention)**.** *When Q, K, V are all derived from the same source X, Definition 0.201 is called* self-attention. *When Q comes from a sequence X but K, V come from a different sequence Y (e.g. encoder outputs in a seq-to-seq model), it is* cross-attention.

Theorems

Theorem 0.203 (Attention as a soft database lookup)**.** *For each query index i, the output row* $\mathrm{Attn}(Q,K,V)_i$ *is a convex combination of the value rows $V_1, \ldots, V_T$.*

Proof. By Chapter 16, $A_{i,:}$ is the image of a softmax, hence $A_{ij} \in (0,1)$ and $\sum_j A_{ij} = 1$. Therefore

$$(AV)_i \;=\; \sum_{j=1}^{T} A_{ij}\, V_j \;\in\; \mathrm{conv}\{V_1, \ldots, V_T\}.$$ □

Theorem 0.204 (Variance of dot products and the $\sqrt{d_k}$ scaling)**.** *Suppose the entries of $Q_{i,:}$ and $K_{j,:}$ are i.i.d., mean zero, variance 1, and the two vectors are independent. Then*

$$\mathbb{E}\big[Q_i \cdot K_j\big] = 0, \qquad \mathrm{Var}\big(Q_i \cdot K_j\big) = d_k.$$

Consequently $\big(QK^\top\big)_{ij}/\sqrt{d_k}$ has unit variance.

Proof. Write

$$S \;=\; Q_i \cdot K_j \;=\; \sum_{k=1}^{d_k} Q_{ik} K_{jk}.$$

By linearity of expectation and independence,

$$\mathbb{E}[S] = \sum_{k=1}^{d_k} \mathbb{E}[Q_{ik}]\, \mathbb{E}[K_{jk}] = 0.$$

For the second moment, expand

$$S^2 = \sum_{k=1}^{d_k} Q_{ik}^2 K_{jk}^2 \; + \; \sum_{k \neq k'} Q_{ik} K_{jk} Q_{ik'} K_{jk'}.$$

The diagonal terms satisfy $\mathbb{E}[Q_{ik}^2 K_{jk}^2] = \mathbb{E}[Q_{ik}^2]\,\mathbb{E}[K_{jk}^2] = 1\cdot 1 = 1$. The off-diagonal terms factor (by independence across k and across Q, K) into products of zero means, hence vanish in expectation. Therefore

$$\mathrm{Var}(S) = \mathbb{E}[S^2] - 0 = d_k. \qquad \square$$

Corollary 0.205 (Saturation without scaling)**.** *Without the $1/\sqrt{d_k}$ factor, the logits feeding softmax have standard deviation $\sqrt{d_k} \to \infty$. The maximum logit dominates and softmax concentrates on a single index $j^\star$. Since $\partial \mathrm{softmax}_i / \partial s_j = \mathrm{softmax}_i(\delta_{ij} - \mathrm{softmax}_j)$, the Jacobian collapses to (approximately) the zero matrix off $j^\star$, vanishing gradients with respect to Q and K.*

Remark 0.206. Scaling by $1/\sqrt{d_k}$ is therefore a *temperature normalization*. It keeps softmax in the well-conditioned regime where its Jacobian has rank $T-1$ and gradients propagate informatively through every position.

Theorem 0.207 (Permutation equivariance)**.** *Let $\pi \in S_T$ be a permutation of the sequence axis, with permutation matrix P_π. Then*

$$\mathrm{Attn}(P_\pi Q,\, P_\pi K,\, P_\pi V) \;=\; P_\pi\, \mathrm{Attn}(Q, K, V).$$

Proof. The pre-softmax logits transform as

$$\big(P_\pi Q\big)\big(P_\pi K\big)^\top / \sqrt{d_k} \;=\; P_\pi \big(QK^\top / \sqrt{d_k}\big) P_\pi^\top.$$

Row-wise softmax acts independently on each row, and the column permutation $P_\pi^\top$ on the right merely reorders the entries within each row, so the softmax permutes consistently:

$$A' := \mathrm{softmax}\big(P_\pi (QK^\top/\sqrt{d_k}) P_\pi^\top\big) = P_\pi A P_\pi^\top.$$

Multiplying by the permuted values,

$$A'(P_\pi V) = P_\pi A P_\pi^\top P_\pi V = P_\pi A V,$$

since $P_\pi^\top P_\pi = I$. $\square$

Remark 0.208. This symmetry is *too strong* for natural language: attention cannot distinguish "the cat sat" from "sat cat the". Chapter 24 fixes this by injecting positional information into X before the projections.

Theorem 0.209 ($O(1)$ gradient depth between any two tokens)**.** *For any pair of indices i, j,*

$$\frac{\partial (AV)_i}{\partial V_j} \;=\; A_{ij}\, I_{d_v},$$

which has spectral norm $A_{ij} \in (0,1)$, independent of $|i-j|$.

Proof. $(AV)_i = \sum_{\ell=1}^{T} A_{i\ell} V_\ell$, so $\partial (AV)_i / \partial V_j = A_{ij} I_{d_v}$. $\square$

Remark 0.210. Compare to Chapter 20: an RNN with spectral norm ρ satisfies $\|\partial h_i / \partial h_j\| \lesssim \rho^{|i-j|}$, exponentially small for $\rho < 1$. Attention replaces this exponential dependence by a single matmul, paying $O(T^2 d_k)$ compute for the matrix A.

Connection to LLMs

Scaled dot-product attention is *the* primitive of every transformer language model. Chapter 22 stacks h heads of it in parallel (multi-head attention); Chapter 23 wires attention into the residual + MLP transformer block; Chapter 24 repairs the permutation symmetry of Theorem 4 with positional encodings. Production-scale variants — Flash Attention (IO-aware tiling), multi-query and grouped-query attention (sharing K, V across heads), sliding-window and linear attention — all preserve the scaled-softmax form derived here. Whenever you call an LLM, every token-to-token interaction inside it is an instance of $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$.

References

[1] A. Vaswani et al. *Attention Is All You Need.* NeurIPS, 2017.

Chapter 22: Multi-head attention: parallel heads as concat-then-project; complexity

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 23: Transformer block: residual + LayerNorm/RMSNorm + FFN + attention; gradient-flow argument

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 24: Positional encoding: sinusoidal derivation, RoPE construction

Stub chapter — will be filled by Phase 2 chapter agent.

Block F — Pre-training

Chapter 25: Causal masking; next-token prediction loss as MLE on the empirical distribution

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 26: Tokenization: BPE algorithm; greedy merge correctness

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 27: Pre-training pipeline: AdamW + warmup + cosine decay + gradient clipping; tiny-GPT training run

Stub chapter — will be filled by Phase 2 chapter agent.

Block G — Post-training

Chapter 28: SFT, RLHF (PPO/GRPO), and DPO; train + post-train a tiny GPT

Stub chapter — will be filled by Phase 2 chapter agent.